P25 J3400: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

Project Details

DETAIL	**VALUE**
Title	24sep20b_WT-Control-White_From-January
Description	This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using
ID	P25
Created by user	Andy Paige

Job Details

DETAIL	**VALUE**
Title	New Job J3400
Description	Enter a description.
ID	J3400
Job type	Non-uniform Refinement
Status	completed
Created by user	Minou Emmad
Created	Tue Jun 09 2026 15:17:59 GMT-0400 (Eastern Daylight Time)
Queued	Tue Jun 09 2026 15:18:19 GMT-0400 (Eastern Daylight Time)
Launched	Tue Jun 09 2026 15:28:50 GMT-0400 (Eastern Daylight Time)
Started	Tue Jun 09 2026 15:28:57 GMT-0400 (Eastern Daylight Time)
Waiting	-
Killed	-
Completed	Tue Jun 09 2026 15:39:35 GMT-0400 (Eastern Daylight Time)
Failed	-
Last accessed	Wed Jun 10 2026 15:57:36 GMT-0400 (Eastern Daylight Time)
Size	3.54 GB

Inputs

particles

J3395.particles_all_classes

BLOB
J3395.particles_all_classes.blob.F

CTF
J3395.particles_all_classes.ctf.F

ALIGNMENTS3D

J3395.particles_all_classes.alignments3D.F

PASSTHROUGH
J3395.particles_all_classes.alignments3D_multi.F

volume

J3395.volume_class_0

MAP
J3395.volume_class_0.map.F

mask

No group connections

Parameters

Particle preprocessing

PARAMETER	**VALUE**	**DEFAULT**	**SPEC**	**ADVANCED**
Window dataset (real-space)	True	X		
Window inner radius	0.85	X		X
Window outer radius	0.99	X		X

Homogeneous Refinement

PARAMETER	**VALUE**	**DEFAULT**	**SPEC**	**ADVANCED**
Symmetry	C1	X		
Symmetry relaxation method	none	X		X
Do symmetry alignment	True	X		X
Re-estimate greyscale level of input reference	True	X		X
Number of extra final passes	0	X		X
Maximum align resolution (A)	None	X		
Initial lowpass resolution (A)	30	X		X
GSFSC split resolution (A)	20	X		X
Force re-do GS split	False	X		X
Adaptive Marginalization	True	X		X
Non-uniform refine enable	True	X		X
Non-uniform filter order	8	X		X
Non-uniform AWF	3	X		X
Enforce non-negativity	False	X		X
Window structure in real space	True	X		X
Skip interpolant premult	True	X		X
Ignore DC component	True	X		X
Ignore tilt	False	X		X

Ignore trefoil	False	X		X
Ignore tetra	False	X		X
Use random ordering of particles	True	X		X
Disable auto batchsize	False	X		X
Batchsize epsilon	0.001	X		X
Batchsize snrfactor	50	X		X
Reset input per-particle scale	True	X		X
Minimize over per-particle scale	False	X		X
Scale min/use start iter	0	X		X
Noise priorw	50	X		X
Noise initw	200	X		X
Noise initial sigma-scale	3	X		X
Initialize noise model from images	False	X		X
Dynamic mask threshold (0-1)	0.2	X		X
Dynamic mask near (A)	6	X		X
Dynamic mask far (A)	14	X		X
Dynamic mask start resolution (A)	12	X		
Dynamic mask use absolute value	False	X		X
Show plots from intermediate steps	True	X		
GPU batch size of images	None	X		X

Defocus Refinement

PARAMETER	**VALUE**	**DEFAULT**	**SPEC**	**ADVANCED**
Optimize per-particle defocus	False	X		X
Num. particles to plot	3	X		
Minimum Fit Res (A)	20	X		
Defocus Search Range (A +/-)	2000	X		
GPU batch size of images	None	X		X

Global CTF Refinement

PARAMETER	**VALUE**	**DEFAULT**	**SPEC**	**ADVANCED**
Optimize per-group CTF params	False	X		X
Num. groups to plot	3	X		
Binning to apply to plots	1	X		X
Minimum Fit Res (A)	10	X		
Fit Tilt	True	X		
Fit Trefoil	True	X		

Fit Spherical Aberration	False	X		
Fit Tetrafoil	False	X		
Fit Anisotropic Mag.	False	X		
GPU batch size of images	None	X		X

Ewald Sphere Correction

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Do EWS correction	False	X		X
EWS curvature sign	negative	X		X

Random Seeds

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Random seed	124843976	X		X

Compute settings

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Cache particle images on SSD	True	X		
Low-Memory Mode	False	X		

Outputs

particles

ALIGNMENTS3D
J3400.particle.alignments3D

CTF
J3400.particle.ctf

BLOB (PASSTHROUGH)
J3400.particle.blob

ALIGNMENTS3D_MULTI (PASSTHROUGH)
J3400.particle.alignments3D_multi

volume

MAP
J3400.volume.blob

MAP_SHARP
J3400.volume.blob

MAP_HALF_A

J3400.volume.blob

MAP_HALF_B
J3400.volume.blob

MASK_REFINE
J3400.volume.blob

MASK_FSC
J3400.volume.blob

MASK_FSC_AUTO
J3400.volume.blob

PRECISION
J3400.volume.blob

mask

MASK_REFINE
J3400.volume.blob

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-09 15:28:50.33 GMT-4] License is valid.
> [2026-06-09 15:28:50.33 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-09 15:28:50.35 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-09 15:28:57.46 GMT-4] [CPU: 78.9 MB]    [Avail: 302.89 GB] Job J3400 Started
> [2026-06-09 15:28:57.50 GMT-4] [CPU: 78.9 MB]    [Avail: 302.85 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-09 15:28:57.52 GMT-4] [CPU: 79.2 MB]    [Avail: 302.83 GB] Working in directory:
                                                                     /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                                                     sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J340
                                                                     0
> [2026-06-09 15:28:57.53 GMT-4] [CPU: 79.2 MB]    [Avail: 302.83 GB] Running on lane default
> [2026-06-09 15:28:57.53 GMT-4] [CPU: 79.2 MB]    [Avail: 302.82 GB] Resources allocated:
> [2026-06-09 15:28:57.53 GMT-4] [CPU: 79.2 MB]    [Avail: 302.82 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-09 15:28:57.54 GMT-4] [CPU: 79.2 MB]    [Avail: 302.82 GB]   CPU   :  [4, 5, 12, 13]
> [2026-06-09 15:28:57.54 GMT-4] [CPU: 79.2 MB]    [Avail: 302.81 GB]   GPU   :  [1]
> [2026-06-09 15:28:57.55 GMT-4] [CPU: 79.2 MB]    [Avail: 302.81 GB]   RAM   :  [2, 7, 8]
> [2026-06-09 15:28:57.55 GMT-4] [CPU: 79.2 MB]    [Avail: 302.80 GB]   SSD   :  True
> [2026-06-09 15:28:57.55 GMT-4] [CPU: 79.2 MB]    [Avail: 302.80 GB] -------------------------------------------------------------
                                                                     -
> [2026-06-09 15:28:57.56 GMT-4] [CPU: 79.2 MB]    [Avail: 302.79 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-09 15:29:08.96 GMT-4] [CPU: 273.6 MB]   [Avail: 300.68 GB] Job ready to run
> [2026-06-09 15:29:08.97 GMT-4] [CPU: 273.6 MB]   [Avail: 300.69 GB] *************************************************************
                                                                     **
> [2026-06-09 15:29:09.20 GMT-4] [CPU: 285.9 MB]   [Avail: 300.81 GB] Using random seed of 124843976
> [2026-06-09 15:29:09.21 GMT-4] [CPU: 285.9 MB]   [Avail: 300.81 GB] Loading a ParticleStack with 3023 items...
> [2026-06-09 15:29:09.22 GMT-4] [CPU: 285.9 MB]   [Avail: 300.80 GB]


                                                                     SSD cache ACTIVE at
                                                                     /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                                                     reserve) (3 TB quota)

                                                                       Cache usage                          Amount

                                                                       Total / Usable          3.58 TiB / 2.73 TiB
                                                                       Used / Free           2.73 TiB / 666.66 MiB
                                                                       Hits / Misses            28.75 GiB / 0.00 B
                                                                       Acquired / Required      28.75 GiB / 0.00 B

                                                                     Progress:
                                                                     [                                                  ]
                                                                     2518/2518 (100%)
                                                                           Elapsed: 0h 00m 03s
                                                                       Active jobs: P25-J3397, P25-J3399, P25-J3400
                                                                     SSD cache complete for 2518 file(s)



> [2026-06-09 15:29:13.19 GMT-4] [CPU: 328.2 MB]   [Avail: 303.82 GB]   Done.
> [2026-06-09 15:29:13.20 GMT-4] [CPU: 328.2 MB]   [Avail: 303.82 GB] Windowing particles
> [2026-06-09 15:29:13.21 GMT-4] [CPU: 329.2 MB]   [Avail: 303.82 GB]   Done.
> [2026-06-09 15:29:13.21 GMT-4] [CPU: 329.2 MB]   [Avail: 303.82 GB] ====== Gold Standard Split ======
> [2026-06-09 15:29:13.21 GMT-4] [CPU: 329.4 MB]   [Avail: 303.82 GB]   Particles have input alignments3D connected, so reusing
                                                                     pre-existing split
> [2026-06-09 15:29:13.22 GMT-4] [CPU: 329.4 MB]   [Avail: 303.82 GB]   Set A is smaller than set B by 7 particles (0.232 percent
                                                                     difference relative to the total dataset).
> [2026-06-09 15:29:13.24 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Split A has 1508 particles
> [2026-06-09 15:29:13.24 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Split B has 1515 particles
> [2026-06-09 15:29:13.24 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB] ====== Refinement ======
> [2026-06-09 15:29:13.25 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Input particles have box size 320
> [2026-06-09 15:29:13.25 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Input particles have pixel size 0.8300
> [2026-06-09 15:29:13.25 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Particles will be zeropadded/truncated to size 320 during
                                                                     alignment
```

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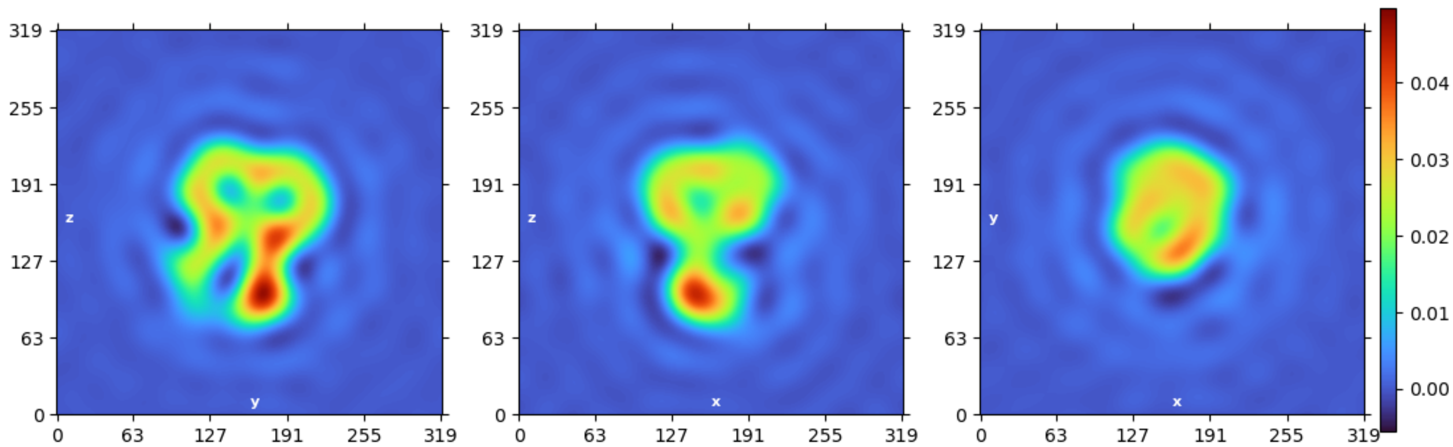
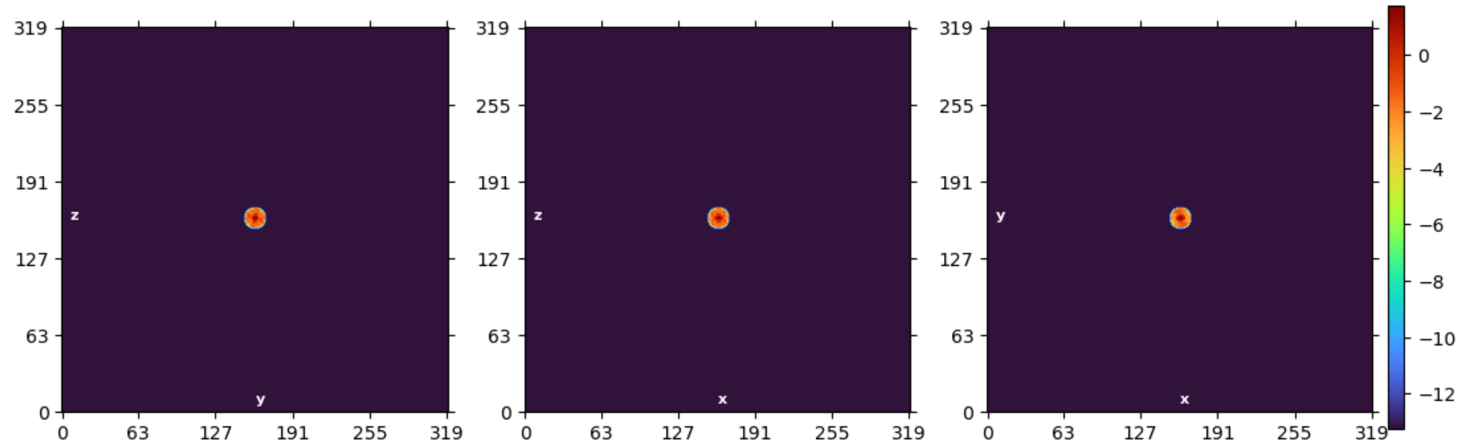
> [2026-06-09 15:29:13.26 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Volume refinement will be done with effective box size 320
> [2026-06-09 15:29:13.26 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Volume refinement will be done with pixel size 0.8300
> [2026-06-09 15:29:13.26 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Particles will be zeropadded/truncated to size 320 during
                                                                       backprojection
> [2026-06-09 15:29:13.26 GMT-4] [CPU: 329.7 MB]   [Avail: 303.81 GB]   Particles will be backprojected with box size 320
> [2026-06-09 15:29:13.27 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Volume will be internally cropped and stored with box size
                                                                       320
> [2026-06-09 15:29:13.27 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Volume will be interpolated with box size 320 (zeropadding
                                                                       factor 1.00)
> [2026-06-09 15:29:13.27 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   DC components of images will be ignored and volume will be
                                                                       floated at each iteration.
> [2026-06-09 15:29:13.28 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Spherical windowing of maps is enabled
> [2026-06-09 15:29:13.28 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Refining with C1 symmetry enforced
> [2026-06-09 15:29:13.28 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB] Refining with pose and shift marginalization during
                                                                       backprojection enabled.
> [2026-06-09 15:29:13.29 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Resetting input per-particle scale factors to 1.0
> [2026-06-09 15:29:13.29 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-09 15:29:13.29 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB] ====== Non-Uniform Refinement ======
> [2026-06-09 15:29:13.29 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Non-Uniform Refinement is enabled.
> [2026-06-09 15:29:13.30 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Using AWF of 3.00.
> [2026-06-09 15:29:13.30 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB]   Using butterworth filter with order 8.
> [2026-06-09 15:29:13.30 GMT-4] [CPU: 329.9 MB]   [Avail: 303.81 GB] ====== Masking ======
> [2026-06-09 15:29:15.64 GMT-4] [CPU: 862.6 MB]   [Avail: 303.05 GB]   No mask input was connected, so dynamic masking will be
                                                                       enabled.
> [2026-06-09 15:29:15.65 GMT-4] [CPU: 862.6 MB]   [Avail: 303.05 GB]   Dynamic mask threshold: 0.2000
> [2026-06-09 15:29:15.65 GMT-4] [CPU: 862.6 MB]   [Avail: 303.04 GB]   Dynamic mask near (A): 6.00
> [2026-06-09 15:29:15.65 GMT-4] [CPU: 862.6 MB]   [Avail: 303.04 GB]   Dynamic mask far  (A): 14.00
> [2026-06-09 15:29:15.65 GMT-4] [CPU: 862.6 MB]   [Avail: 303.04 GB] ====== Initial Model ======
> [2026-06-09 15:29:15.66 GMT-4] [CPU: 862.6 MB]   [Avail: 303.03 GB]   Resampling initial model to specified volume representation
                                                                       size and pixel-size...
> [2026-06-09 15:29:18.81 GMT-4] [CPU: 1.24 GB]    [Avail: 301.59 GB]   Estimating scale of initial reference.
> [2026-06-09 15:29:23.69 GMT-4] [CPU: 77.4 MB]    [Avail: 300.64 GB] WARNING: io_uring support disabled (not supported by kernel),
                                                                       I/O performance may degrade
> [2026-06-09 15:29:24.81 GMT-4] [CPU: 1.45 GB]    [Avail: 303.01 GB]   Rescaling initial reference by a factor of 1.426
> [2026-06-09 15:29:26.52 GMT-4] [CPU: 1.45 GB]    [Avail: 302.04 GB]   Estimating scale of initial reference.
> [2026-06-09 15:29:29.38 GMT-4] [CPU: 1.41 GB]    [Avail: 299.69 GB]   Rescaling initial reference by a factor of 1.001
> [2026-06-09 15:29:30.89 GMT-4] [CPU: 1.41 GB]    [Avail: 300.49 GB]   Estimating scale of initial reference.
> [2026-06-09 15:29:33.55 GMT-4] [CPU: 1.41 GB]    [Avail: 300.50 GB]   Rescaling initial reference by a factor of 0.999
> [2026-06-09 15:29:36.54 GMT-4] Initial Real Space Slices
```

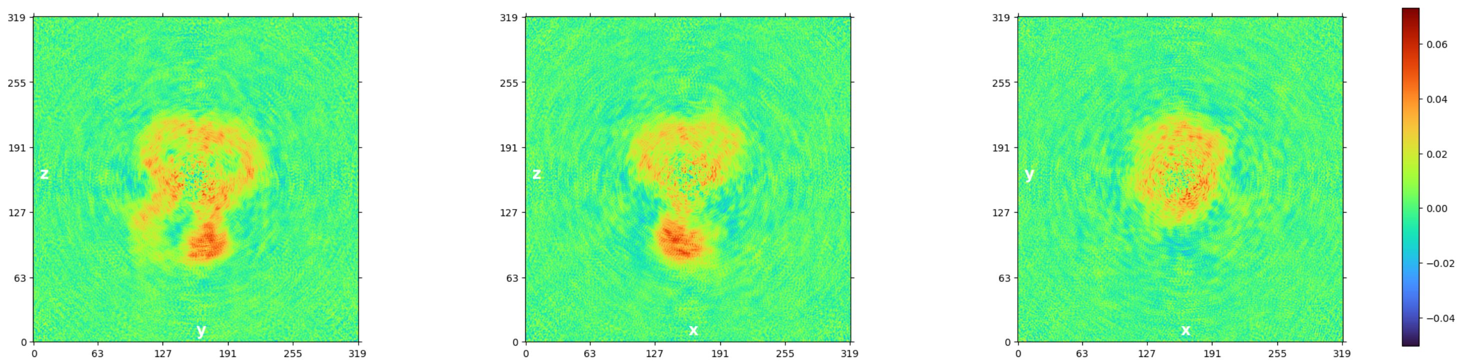


```
> [2026-06-09 15:29:38.35 GMT-4] Initial Fourier Space Slices
```

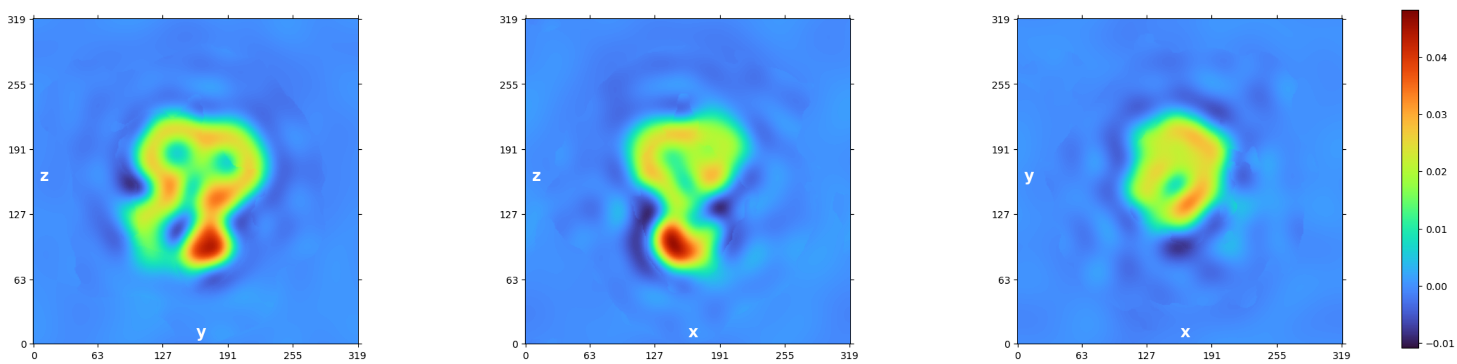



```
> [2026-06-09 15:29:38.35 GMT-4] [CPU: 1.46 GB]    [Avail: 300.14 GB] ====== Starting Refinement Iterations ======
> [2026-06-09 15:29:38.35 GMT-4] [CPU: 1.46 GB]    [Avail: 300.14 GB] ---------------------------- Start Iteration 0
> [2026-06-09 15:29:38.36 GMT-4] [CPU: 1.46 GB]    [Avail: 300.14 GB]   Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-09 15:29:38.36 GMT-4] [CPU: 1.46 GB]    [Avail: 300.14 GB]  Auto batchsize: 1515 in each split
> [2026-06-09 15:29:42.15 GMT-4] [CPU: 1.97 GB]    [Avail: 300.34 GB] -- THR 1 BATCH 500 NUM 500 TOTAL 1.2075967 ELAPSED 14.760472 --

> [2026-06-09 15:29:59.08 GMT-4] [CPU: 3.19 GB]    [Avail: 299.10 GB]   Processed 3023.000 images in 16.931s.
> [2026-06-09 15:30:03.22 GMT-4] [CPU: 3.52 GB]    [Avail: 298.33 GB]   Computing FSCs...
> [2026-06-09 15:30:03.22 GMT-4] [CPU: 3.52 GB]    [Avail: 298.33 GB]   Using full box size 320, downsampled box size 160, with low memory mode disabled.
> [2026-06-09 15:30:03.23 GMT-4] [CPU: 3.52 GB]    [Avail: 298.33 GB]   Computing FFTs on GPU.
> [2026-06-09 15:30:05.85 GMT-4] [CPU: 3.77 GB]    [Avail: 297.97 GB]     Done in 2.625s
> [2026-06-09 15:30:05.85 GMT-4] [CPU: 3.77 GB]    [Avail: 297.97 GB]   Computing cFSCs...
> [2026-06-09 15:30:07.83 GMT-4] [CPU: 3.77 GB]    [Avail: 298.19 GB]     Done in 1.970s
> [2026-06-09 15:30:07.83 GMT-4] [CPU: 3.77 GB]    [Avail: 298.23 GB]   Using Filter Radius 21.635 (12.276A) | Previous: 8.853 (30.000A)
> [2026-06-09 15:30:13.71 GMT-4] [CPU: 4.84 GB]    [Avail: 298.43 GB]   Non-uniform regularization with compute option: GPU
> [2026-06-09 15:30:13.71 GMT-4] [CPU: 4.84 GB]    [Avail: 298.43 GB]   Running local cross validation for A ...
> [2026-06-09 15:30:41.52 GMT-4] [CPU: 7.54 GB]    [Avail: 293.48 GB]   Local cross validation A done in 27.805s
> [2026-06-09 15:30:42.83 GMT-4] FSC Filtered Side A
```

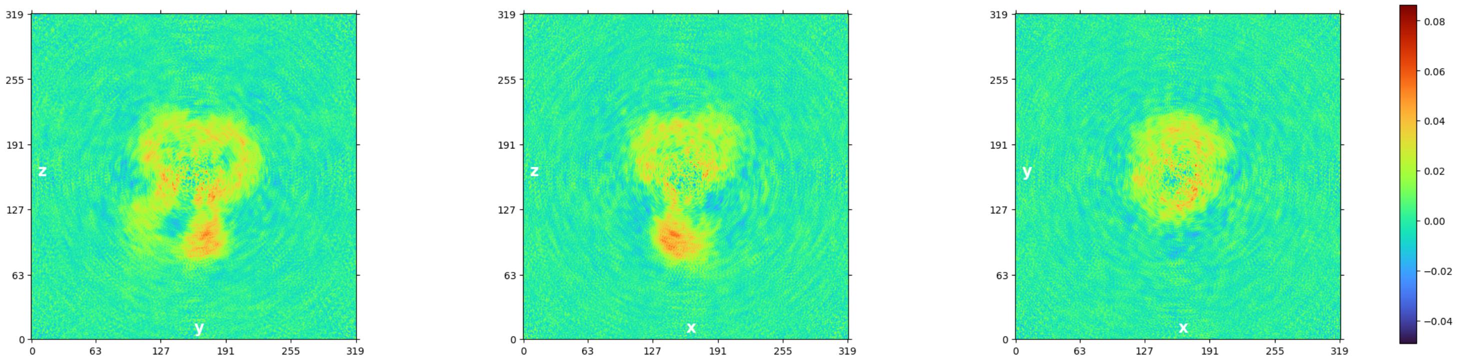



```
> [2026-06-09 15:30:43.80 GMT-4] CV Filtered Side A
```

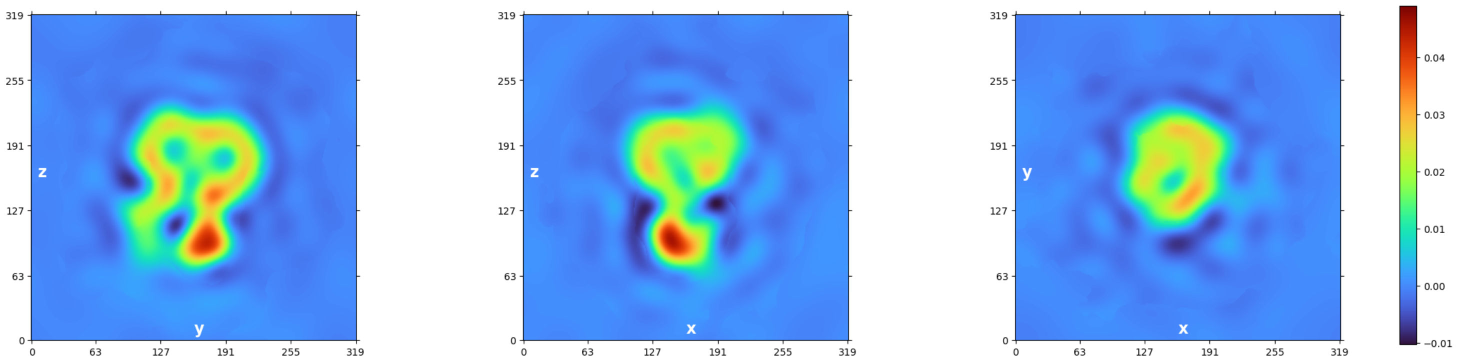



```
> [2026-06-09 15:30:43.80 GMT-4] [CPU: 7.55 GB]    [Avail: 293.83 GB]   Running local cross validation for B ...
> [2026-06-09 15:31:05.59 GMT-4] [CPU: 7.83 GB]    [Avail: 300.77 GB]   Local cross validation B done in 21.787s
```

> [2026-06-09 15:31:06.87 GMT-4] FSC Filtered Side B

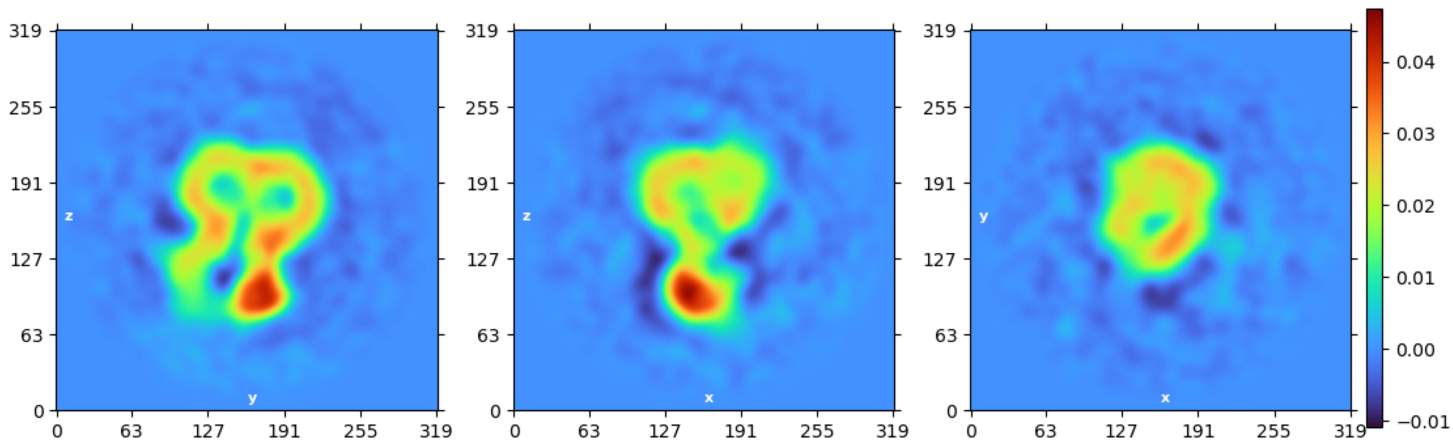


> [2026-06-09 15:31:08.15 GMT-4] CV Filtered Side B

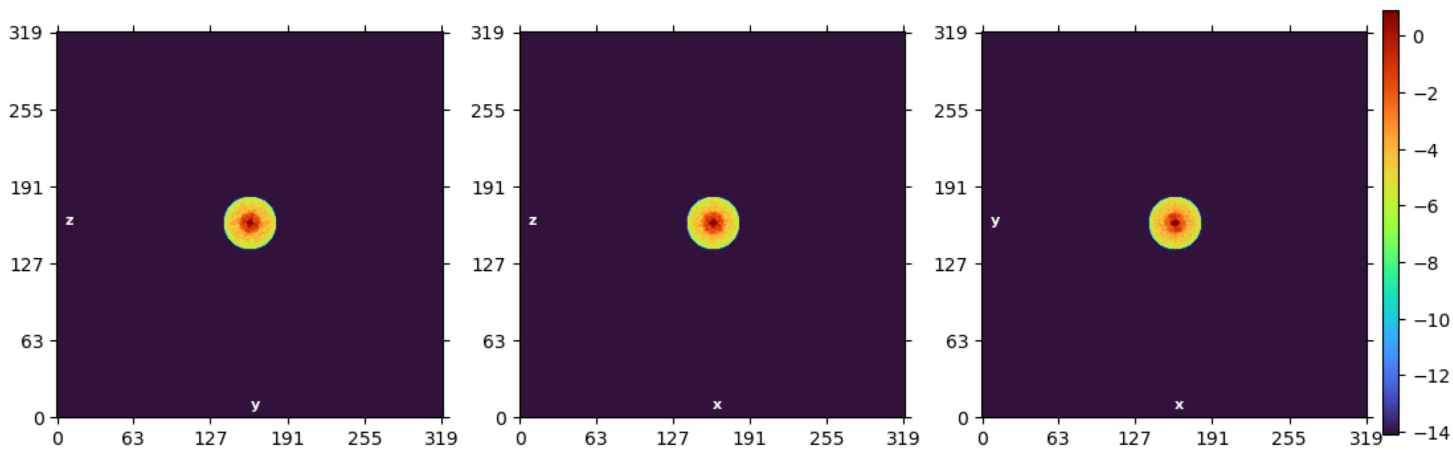


> [2026-06-09 15:31:14.73 GMT-4] [CPU: 8.05 GB] [Avail: 298.55 GB] Plotting..

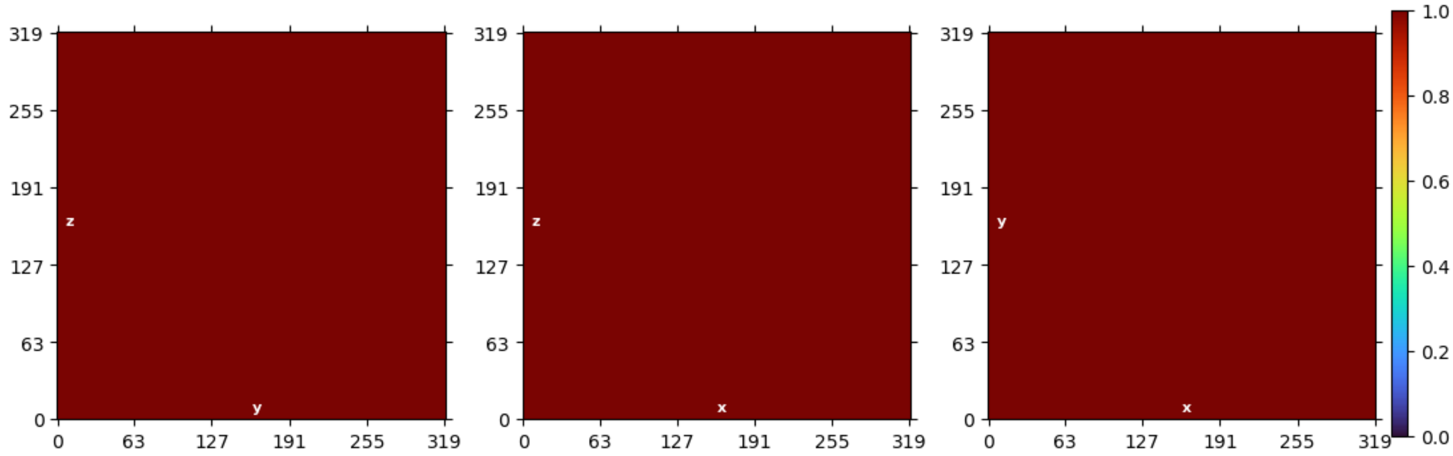
> [2026-06-09 15:31:22.07 GMT-4] Real Space Slices Iteration 000



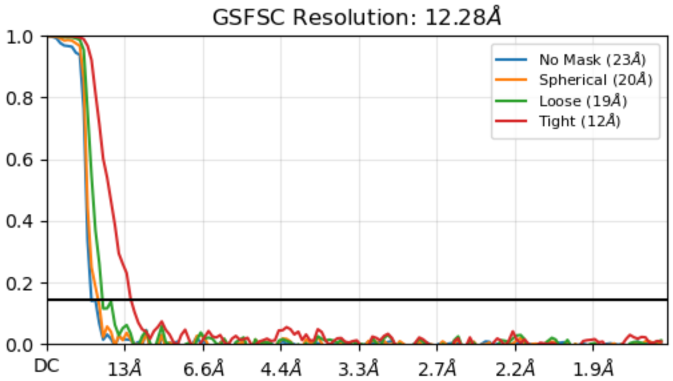
> [2026-06-09 15:31:23.56 GMT-4] Fourier Space Slices Iteration 000



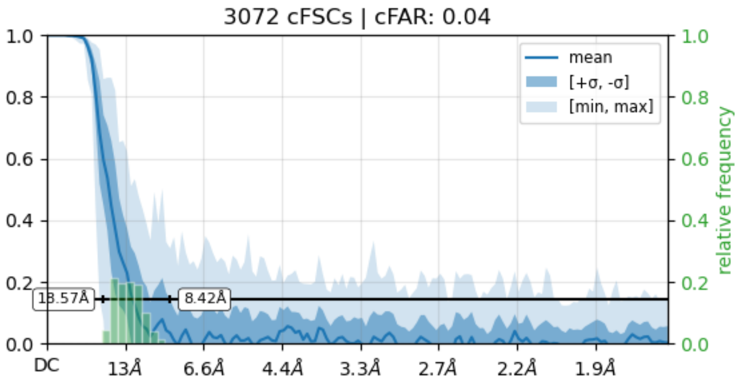
> [2026-06-09 15:31:24.60 GMT-4] Real Space Mask Slices Iteration 000



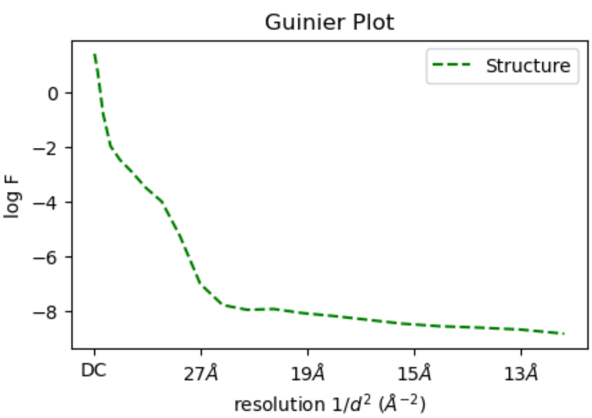
> [2026-06-09 15:31:25.28 GMT-4] FSC Iteration 000



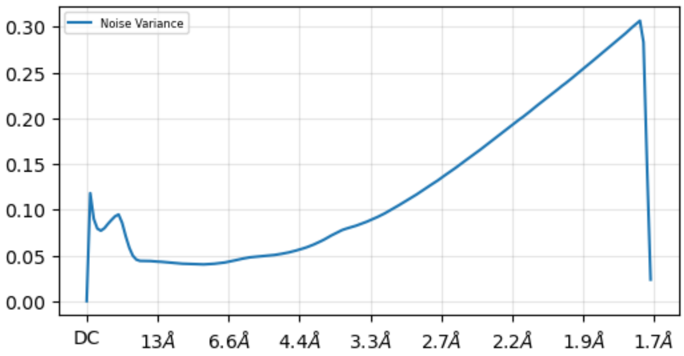
> [2026-06-09 15:31:26.40 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



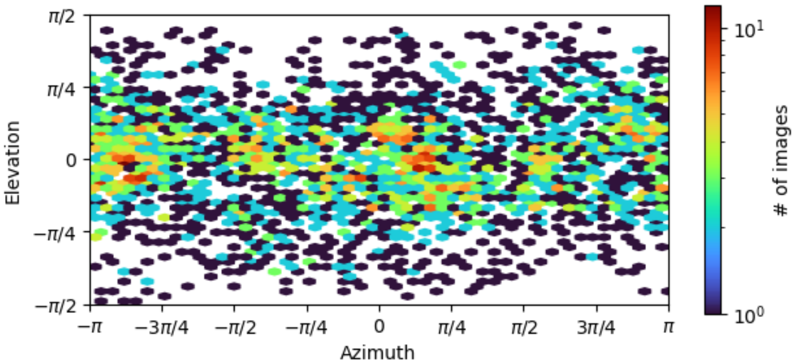
> [2026-06-09 15:31:31.57 GMT-4] Guinier Plot Iteration 000



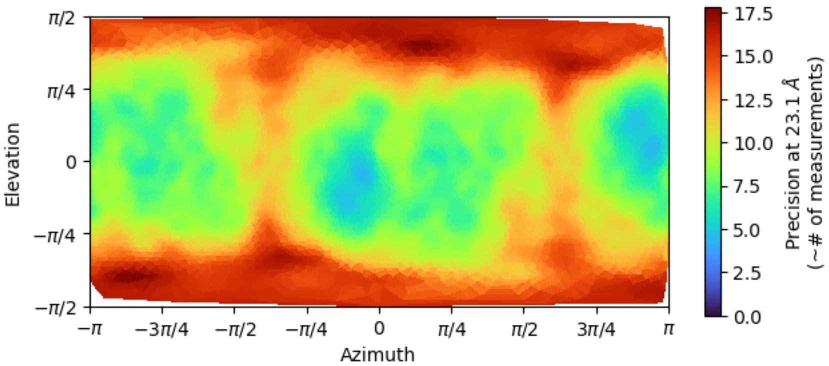
> [2026-06-09 15:31:31.90 GMT-4] Noise Model Iteration 000



> [2026-06-09 15:31:32.55 GMT-4] Viewing Direction Distribution Iteration 000



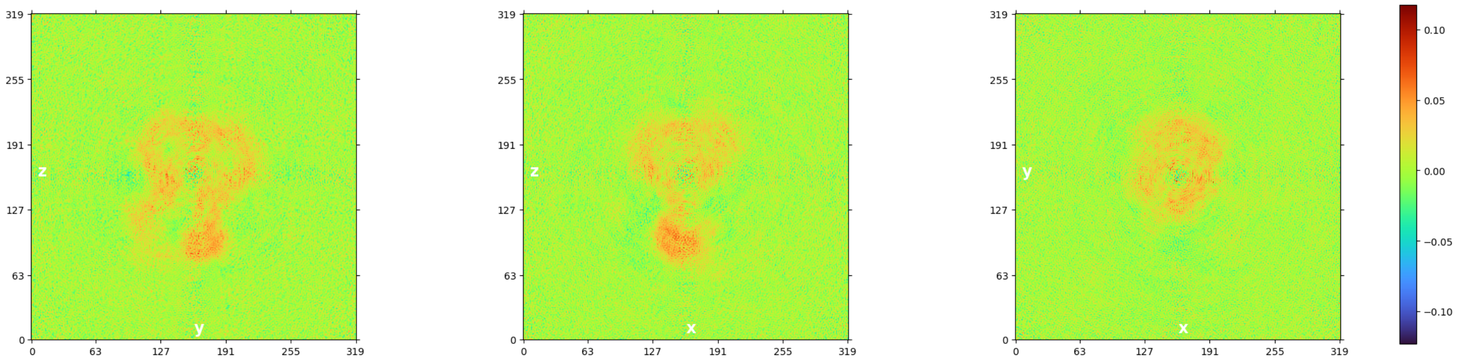
> [2026-06-09 15:31:33.94 GMT-4] Posterior Precision Directional Distribution Iteration 000



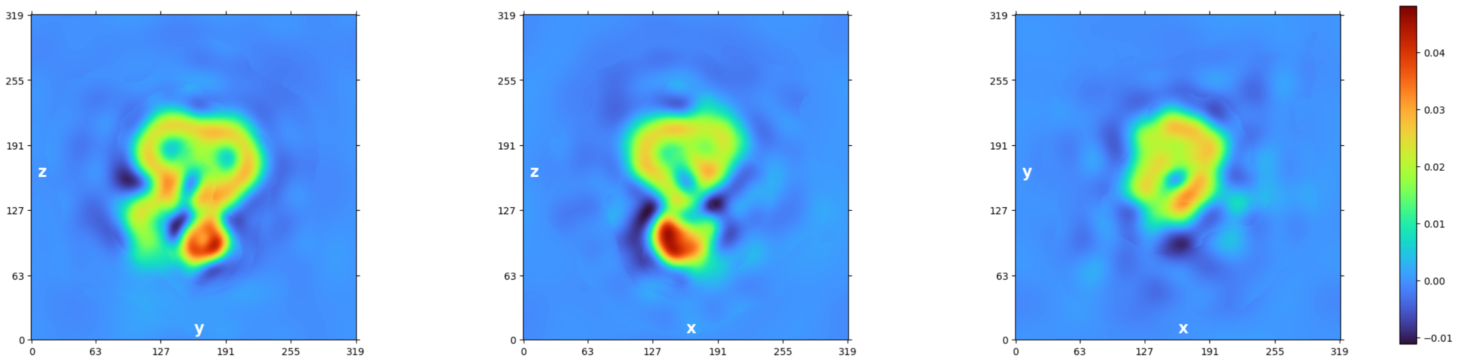

```
> [2026-06-09 15:31:33.95 GMT-4] [CPU: 8.49 GB]    [Avail: 297.09 GB]     Done in 19.210s.
> [2026-06-09 15:31:33.95 GMT-4] [CPU: 8.49 GB]    [Avail: 297.08 GB]   Outputting files..
> [2026-06-09 15:31:49.35 GMT-4] [CPU: 8.92 GB]    [Avail: 293.01 GB]     Done in 15.395s.
> [2026-06-09 15:31:49.35 GMT-4] [CPU: 8.92 GB]    [Avail: 293.00 GB] Done iteration 0 in 130.995s. Total time so far 160.358s
> [2026-06-09 15:31:49.99 GMT-4] [CPU: 8.92 GB]    [Avail: 292.95 GB] ----------------------------- Start Iteration 1
> [2026-06-09 15:31:50.00 GMT-4] [CPU: 8.92 GB]    [Avail: 292.95 GB]   Using Max Alignment Radius 21.635 (12.276A)
> [2026-06-09 15:31:50.00 GMT-4] [CPU: 8.92 GB]    [Avail: 292.95 GB]  Auto batchsize: 1515 in each split
> [2026-06-09 15:31:53.71 GMT-4] [CPU: 8.92 GB]    [Avail: 293.22 GB] -- THR 0 BATCH 500 NUM 500 TOTAL 0.8988547 ELAPSED 13.114439
--

> [2026-06-09 15:32:09.21 GMT-4] [CPU: 9.77 GB]    [Avail: 292.40 GB]   Processed 3023.000 images in 15.496s.
> [2026-06-09 15:32:12.24 GMT-4] [CPU: 9.78 GB]    [Avail: 292.59 GB]   Computing FSCs...
> [2026-06-09 15:32:12.25 GMT-4] [CPU: 9.78 GB]    [Avail: 292.58 GB]   Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:32:12.25 GMT-4] [CPU: 9.78 GB]    [Avail: 292.57 GB]   Computing FFTs on GPU.
> [2026-06-09 15:32:14.13 GMT-4] [CPU: 9.78 GB]    [Avail: 291.50 GB]     Done in 1.887s
> [2026-06-09 15:32:14.14 GMT-4] [CPU: 9.78 GB]    [Avail: 291.50 GB]   Computing cFSCs...
> [2026-06-09 15:32:16.17 GMT-4] [CPU: 9.78 GB]    [Avail: 291.42 GB]     Done in 2.035s
> [2026-06-09 15:32:16.18 GMT-4] [CPU: 9.78 GB]    [Avail: 291.42 GB]   Using Filter Radius 25.986 (10.221A) | Previous: 21.635
(12.276A)
> [2026-06-09 15:32:22.24 GMT-4] [CPU: 9.02 GB]    [Avail: 291.62 GB]   Non-uniform regularization with compute option: GPU
> [2026-06-09 15:32:22.25 GMT-4] [CPU: 9.02 GB]    [Avail: 291.62 GB]   Running local cross validation for A ...
> [2026-06-09 15:32:44.45 GMT-4] [CPU: 9.27 GB]    [Avail: 290.57 GB]   Local cross validation A done in 22.198s
```

> [2026-06-09 15:32:45.66 GMT-4] FSC Filtered Side A



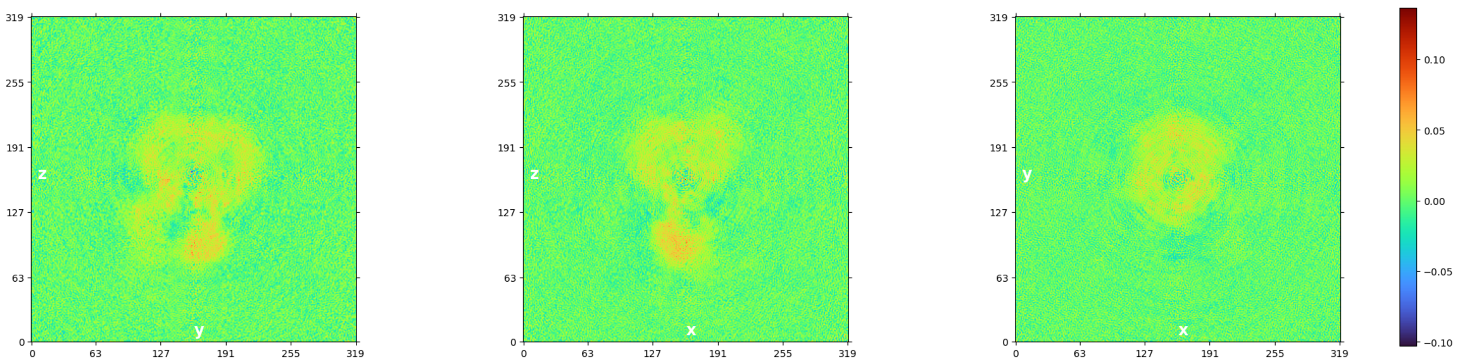
> [2026-06-09 15:32:46.61 GMT-4] CV Filtered Side A



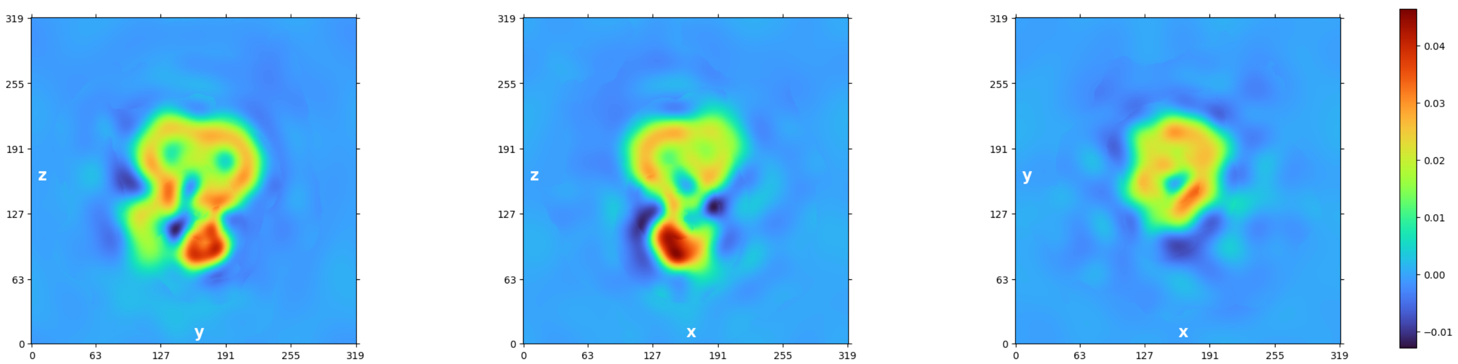
> [2026-06-09 15:32:46.62 GMT-4] [CPU: 9.27 GB] [Avail: 290.23 GB] Running local cross validation for B ...

> [2026-06-09 15:33:09.54 GMT-4] [CPU: 9.52 GB] [Avail: 286.84 GB] Local cross validation B done in 22.917s

> [2026-06-09 15:33:11.12 GMT-4] FSC Filtered Side B

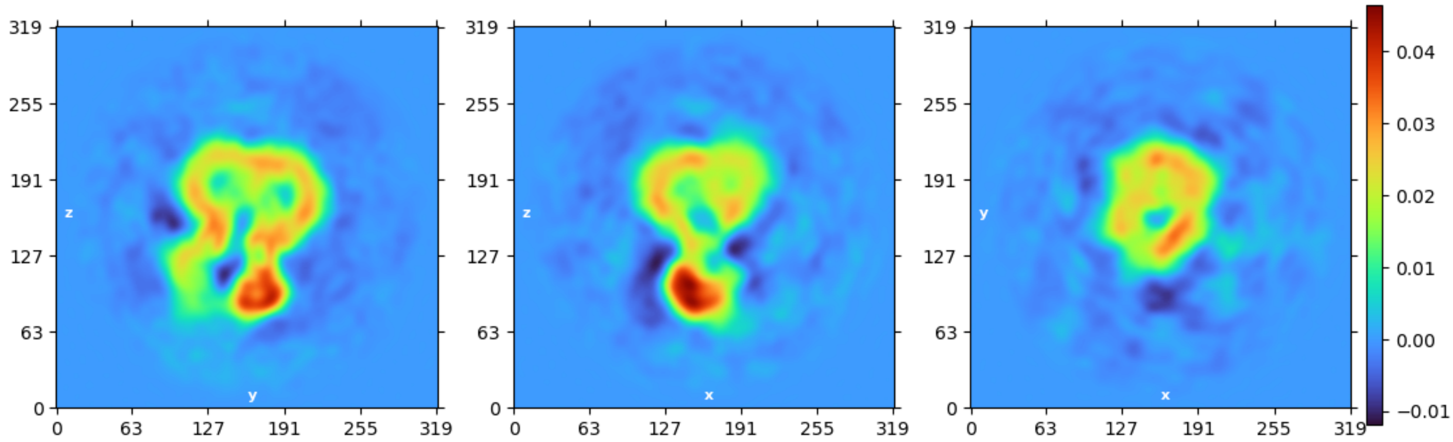


> [2026-06-09 15:33:12.31 GMT-4] CV Filtered Side B

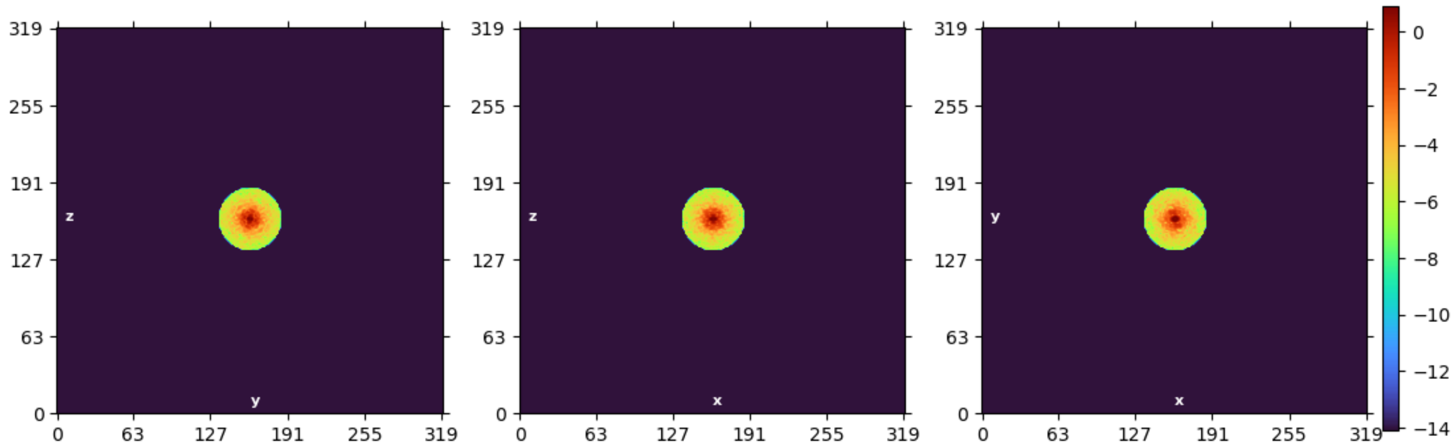


> [2026-06-09 15:33:18.00 GMT-4] [CPU: 9.40 GB] [Avail: 288.52 GB] Plotting..

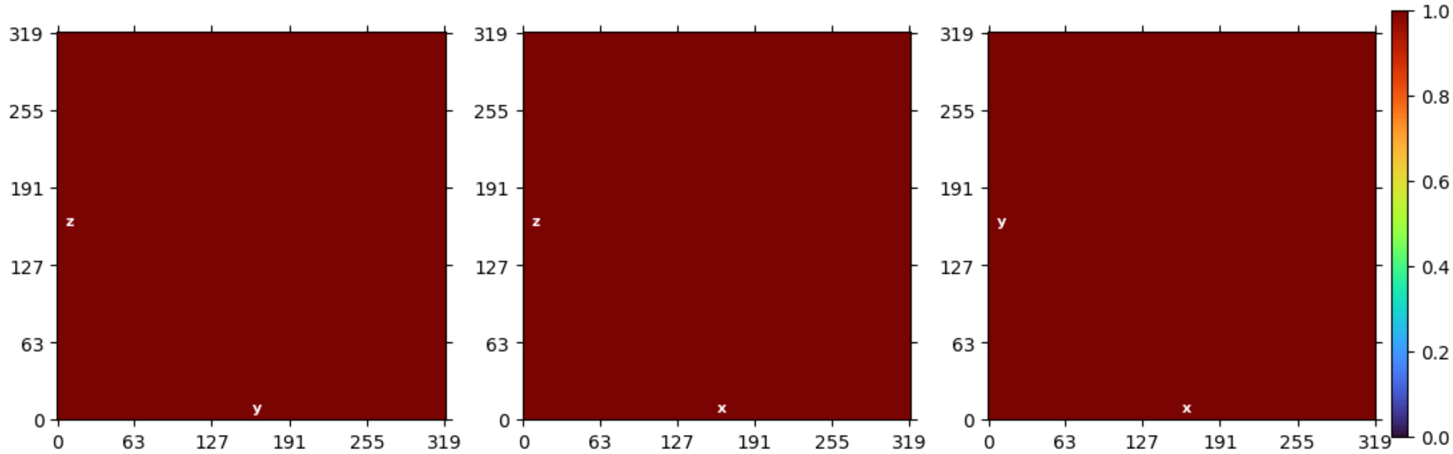
> [2026-06-09 15:33:24.18 GMT-4] Real Space Slices Iteration 001



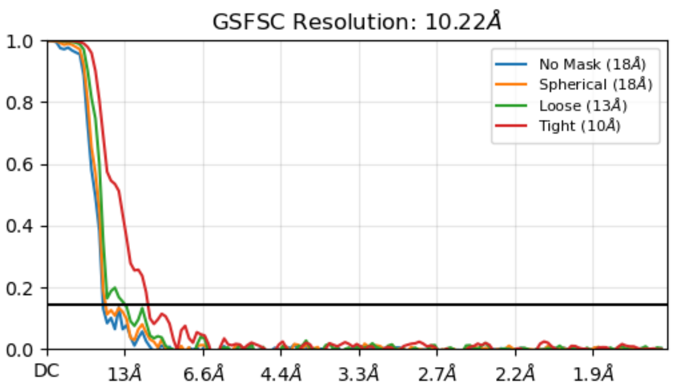
> [2026-06-09 15:33:25.51 GMT-4] Fourier Space Slices Iteration 001



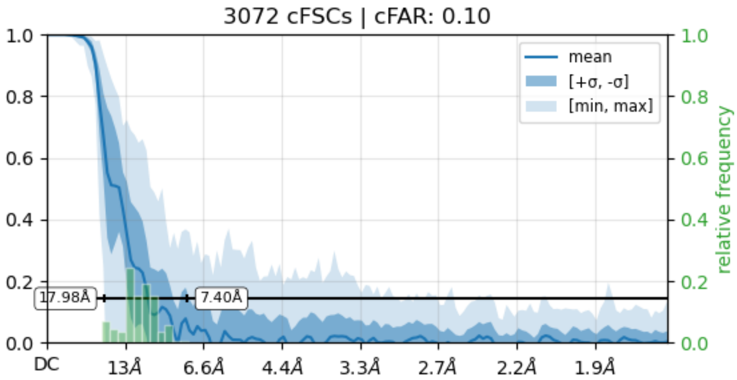
> [2026-06-09 15:33:26.54 GMT-4] Real Space Mask Slices Iteration 001



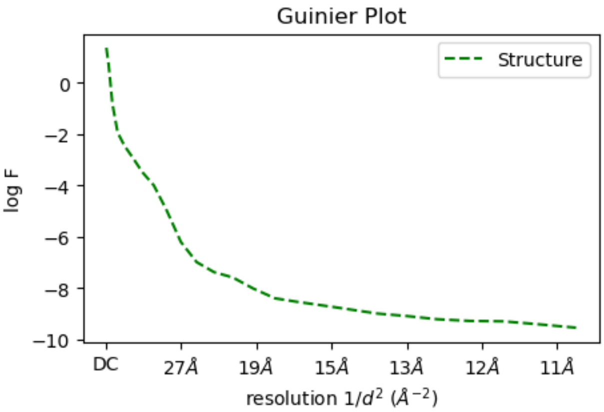
> [2026-06-09 15:33:27.15 GMT-4] FSC Iteration 001



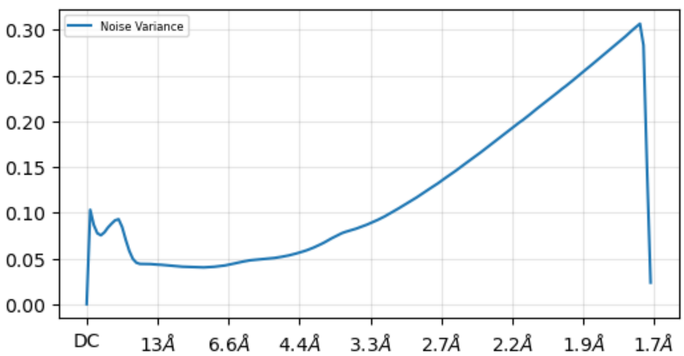
> [2026-06-09 15:33:27.92 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



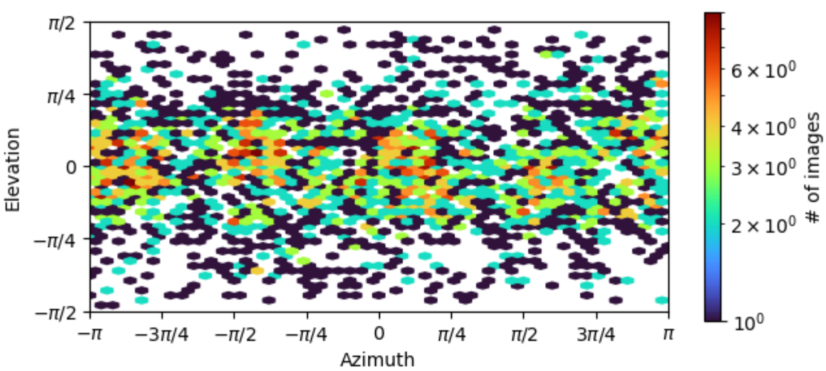
> [2026-06-09 15:33:33.58 GMT-4] Guinier Plot Iteration 001



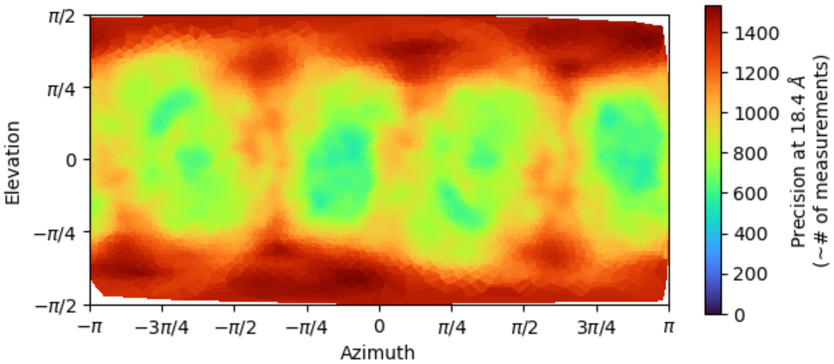
> [2026-06-09 15:33:34.04 GMT-4] Noise Model Iteration 001



> [2026-06-09 15:33:35.01 GMT-4] Viewing Direction Distribution Iteration 001

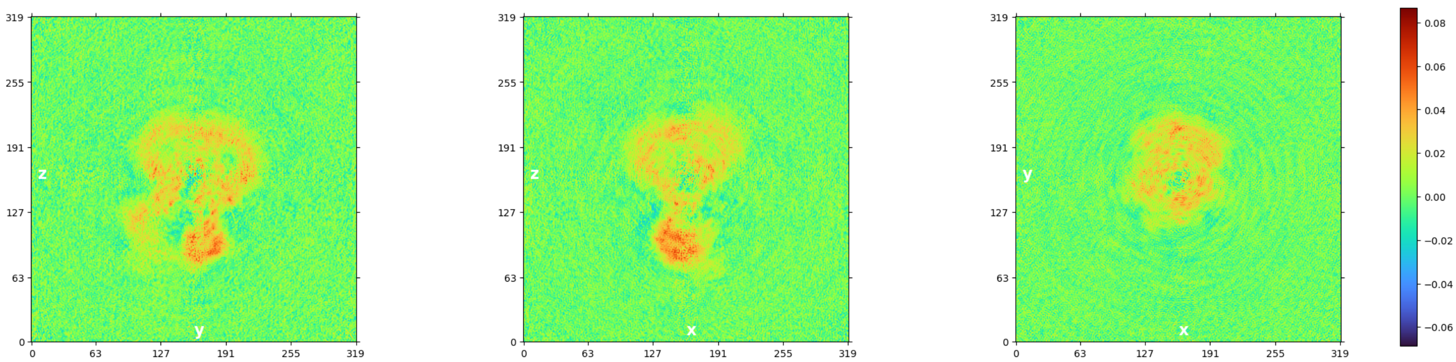


```
> [2026-06-09 15:33:36.78 GMT-4] Posterior Precision Directional Distribution Iteration 001
```

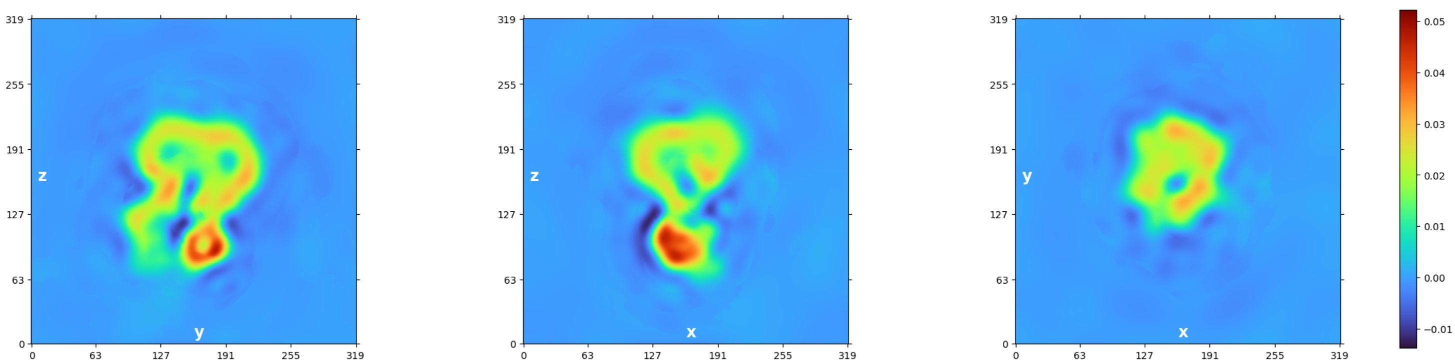



```
> [2026-06-09 15:33:36.79 GMT-4] [CPU: 9.46 GB]    [Avail: 285.77 GB]     Done in 18.787s.
> [2026-06-09 15:33:36.79 GMT-4] [CPU: 9.46 GB]    [Avail: 285.77 GB]   Outputting files..
> [2026-06-09 15:33:51.36 GMT-4] [CPU: 9.37 GB]    [Avail: 288.26 GB]     Done in 14.566s.
> [2026-06-09 15:33:51.36 GMT-4] [CPU: 9.37 GB]    [Avail: 288.25 GB] Done iteration 1 in 121.368s. Total time so far 282.370s
> [2026-06-09 15:33:52.01 GMT-4] [CPU: 9.37 GB]    [Avail: 287.37 GB] ----------------------------- Start Iteration 2
> [2026-06-09 15:33:52.01 GMT-4] [CPU: 9.37 GB]    [Avail: 287.37 GB]   Using Max Alignment Radius 25.986 (10.221A)
> [2026-06-09 15:33:52.01 GMT-4] [CPU: 9.37 GB]    [Avail: 287.36 GB]  Auto batchsize: 1515 in each split
> [2026-06-09 15:33:52.02 GMT-4] [CPU: 9.37 GB]    [Avail: 287.36 GB]   Using dynamic mask.
> [2026-06-09 15:34:20.37 GMT-4] [CPU: 9.87 GB]    [Avail: 284.65 GB] -- THR 1 BATCH 500 NUM 500 TOTAL 1.3199746 ELAPSED 11.833678
--

> [2026-06-09 15:34:34.56 GMT-4] [CPU: 10.72 GB]   [Avail: 288.21 GB]   Processed 3023.000 images in 14.189s.
> [2026-06-09 15:34:37.72 GMT-4] [CPU: 10.72 GB]   [Avail: 288.71 GB]   Computing FSCs...
> [2026-06-09 15:34:37.73 GMT-4] [CPU: 10.72 GB]   [Avail: 288.71 GB]   Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:34:37.73 GMT-4] [CPU: 10.72 GB]   [Avail: 288.71 GB]   Computing FFTs on GPU.
> [2026-06-09 15:34:39.47 GMT-4] [CPU: 10.72 GB]   [Avail: 288.39 GB]     Done in 1.740s
> [2026-06-09 15:34:39.47 GMT-4] [CPU: 10.72 GB]   [Avail: 288.39 GB]   Computing cFSCs...
> [2026-06-09 15:34:41.44 GMT-4] [CPU: 10.72 GB]   [Avail: 286.19 GB]     Done in 1.968s
> [2026-06-09 15:34:41.45 GMT-4] [CPU: 10.72 GB]   [Avail: 286.19 GB]   Using Filter Radius 29.506 (9.002A) | Previous: 25.986
(10.221A)
> [2026-06-09 15:34:48.11 GMT-4] [CPU: 9.97 GB]    [Avail: 293.59 GB]   Non-uniform regularization with compute option: GPU
> [2026-06-09 15:34:48.12 GMT-4] [CPU: 9.97 GB]    [Avail: 293.59 GB]   Running local cross validation for A ...
> [2026-06-09 15:35:11.27 GMT-4] [CPU: 10.22 GB]   [Avail: 299.05 GB]   Local cross validation A done in 23.149s
> [2026-06-09 15:35:12.56 GMT-4] FSC Filtered Side A
```



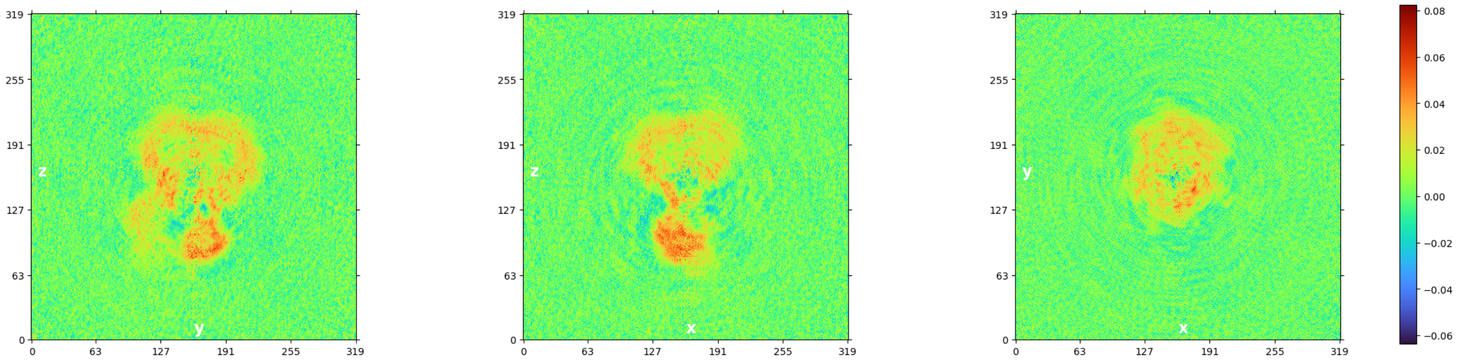

```
> [2026-06-09 15:35:13.64 GMT-4] CV Filtered Side A
```



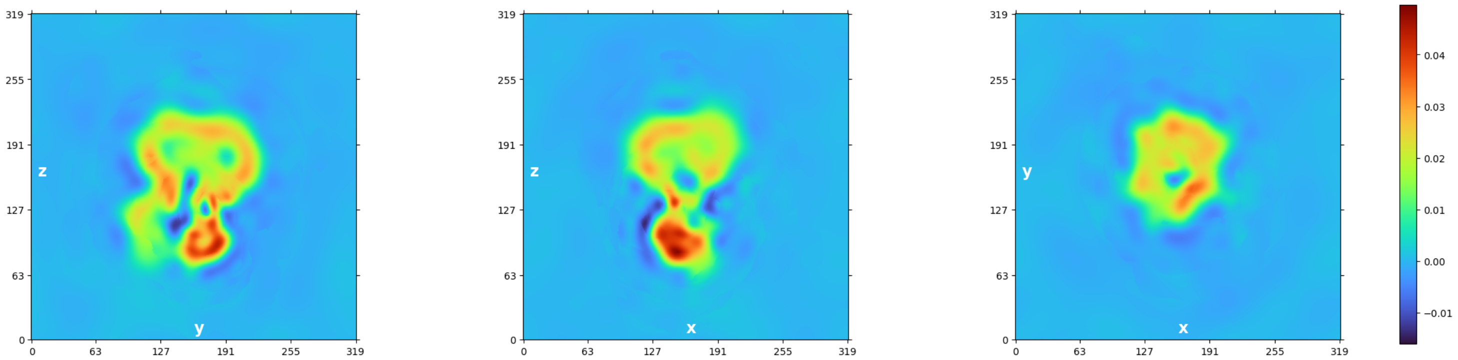

```
> [2026-06-09 15:35:13.64 GMT-4] [CPU: 10.22 GB]   [Avail: 299.52 GB]   Running local cross validation for B ...
```

> [2026-06-09 15:35:35.12 GMT-4] [CPU: 10.47 GB] [Avail: 297.33 GB] Local cross validation B done in 21.476s

> [2026-06-09 15:35:36.41 GMT-4] FSC Filtered Side B



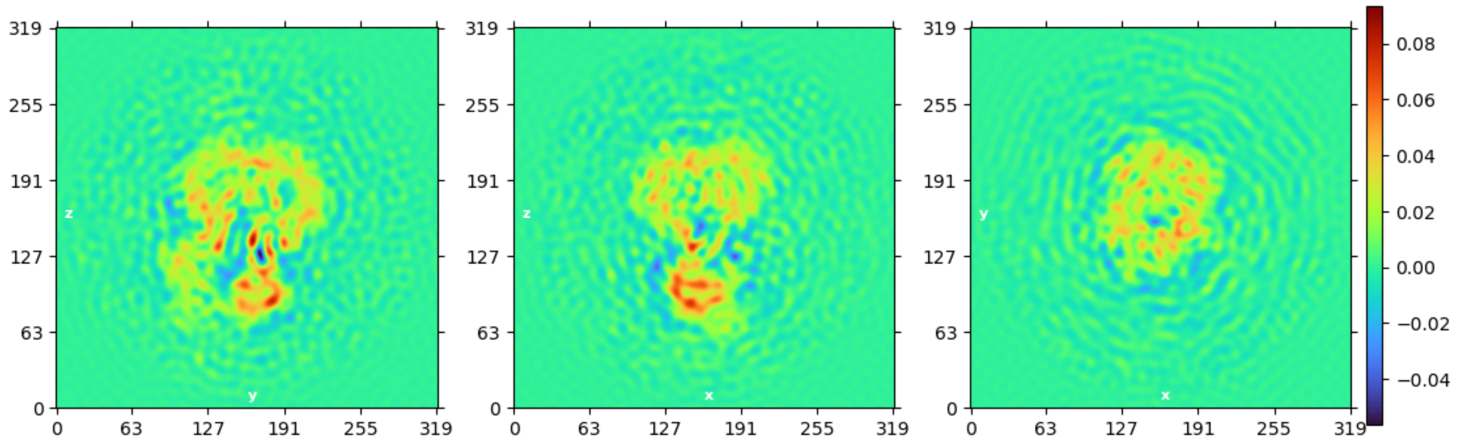
> [2026-06-09 15:35:37.56 GMT-4] CV Filtered Side B



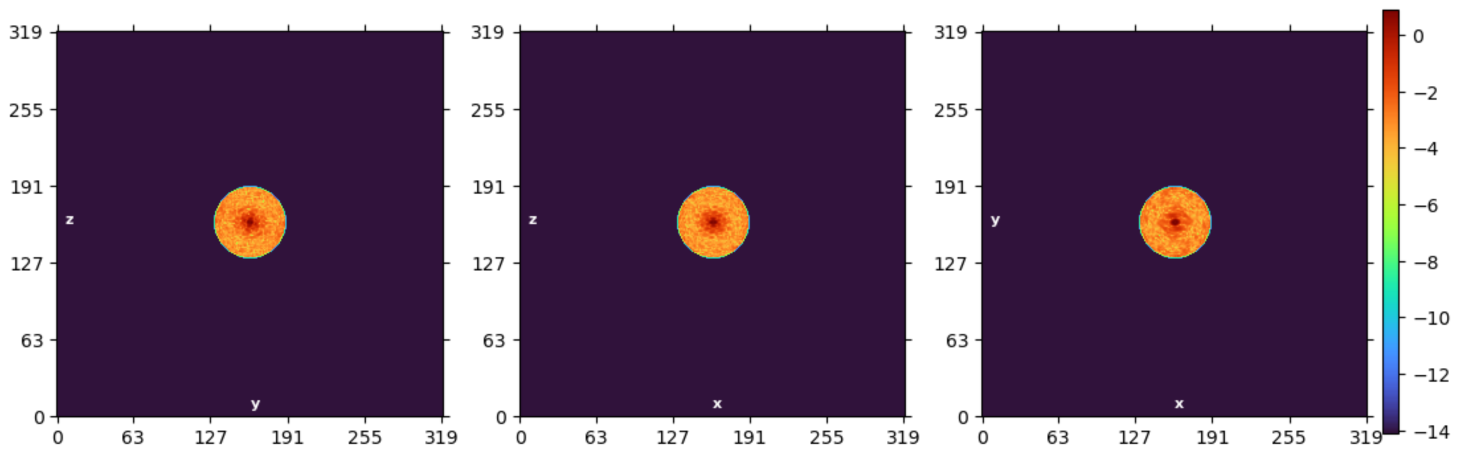
> [2026-06-09 15:35:47.52 GMT-4] [CPU: 10.69 GB] [Avail: 296.58 GB] Estimated Bfactor: -1147.2

> [2026-06-09 15:35:47.53 GMT-4] [CPU: 10.69 GB] [Avail: 296.58 GB] Plotting..

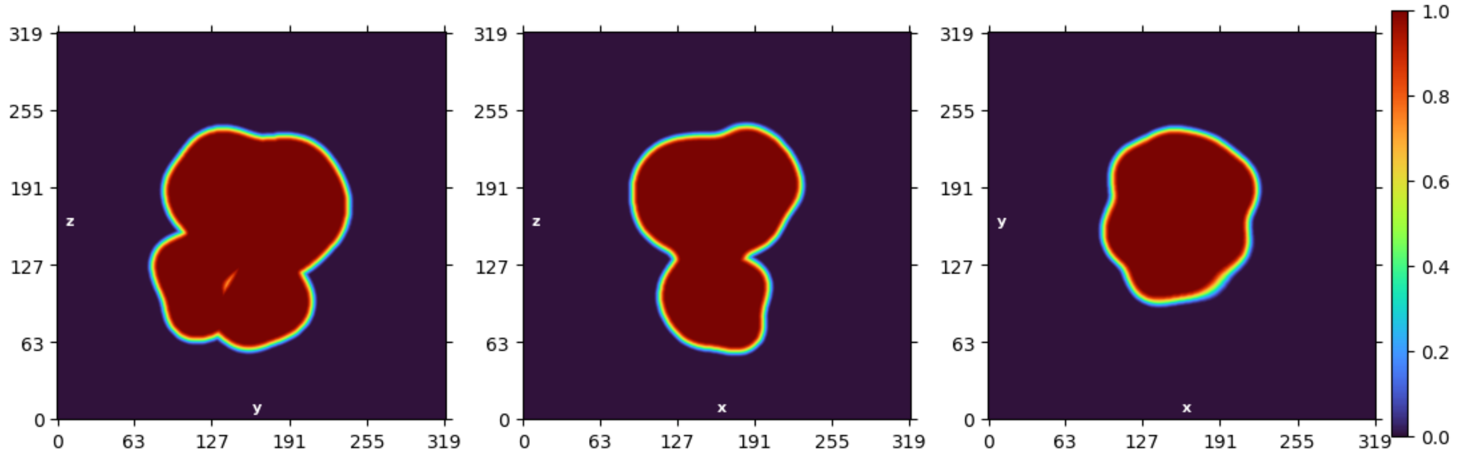
> [2026-06-09 15:35:53.65 GMT-4] Real Space Slices Iteration 002



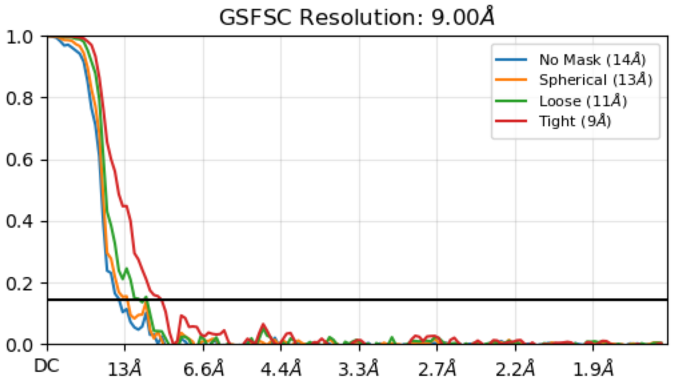
> [2026-06-09 15:35:55.09 GMT-4] Fourier Space Slices Iteration 002



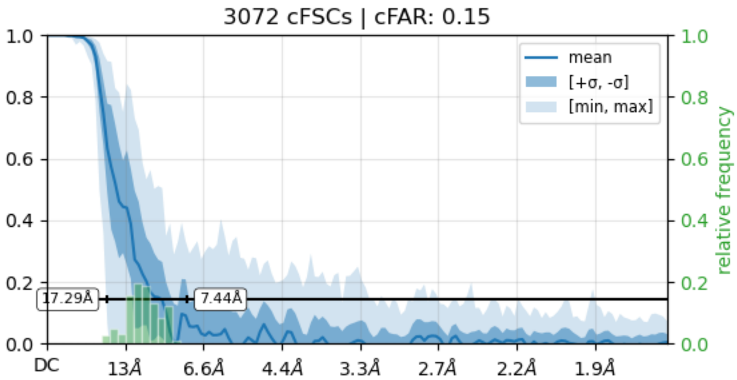
> [2026-06-09 15:35:56.91 GMT-4] Real Space Mask Slices Iteration 002



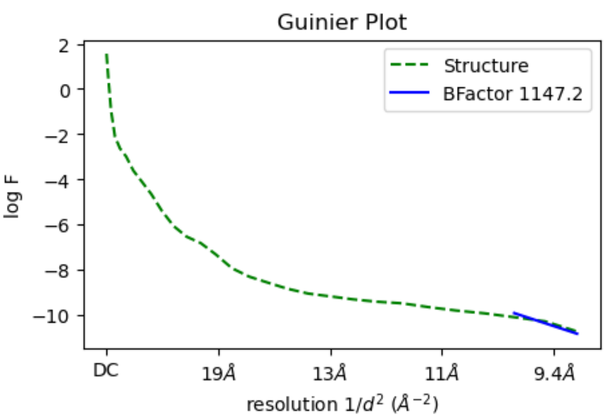
> [2026-06-09 15:35:57.41 GMT-4] FSC Iteration 002



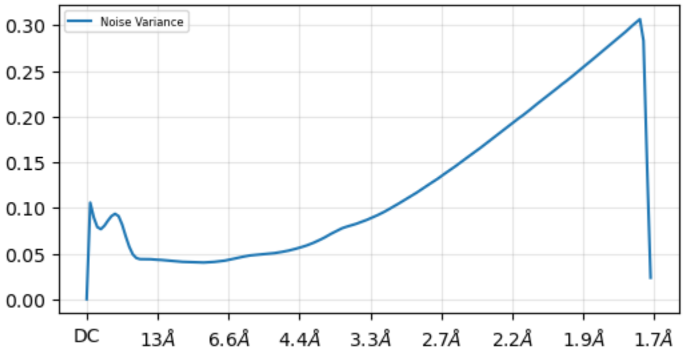
> [2026-06-09 15:35:58.10 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with tight mask



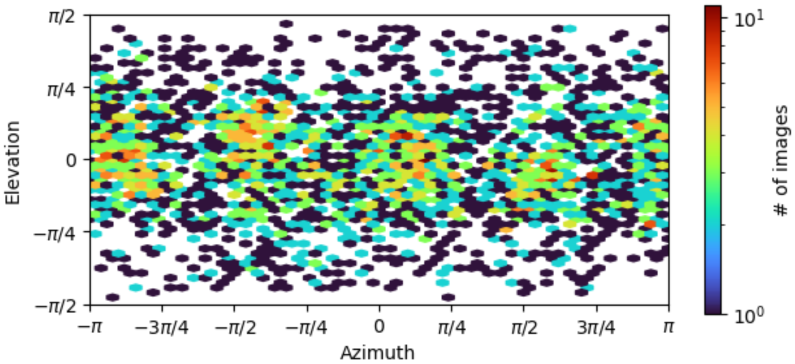
> [2026-06-09 15:36:03.64 GMT-4] Guinier Plot Iteration 002



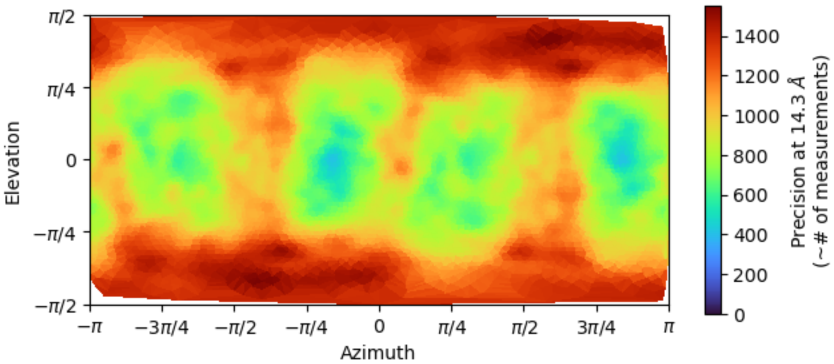
> [2026-06-09 15:36:04.00 GMT-4] Noise Model Iteration 002



> [2026-06-09 15:36:04.72 GMT-4] Viewing Direction Distribution Iteration 002

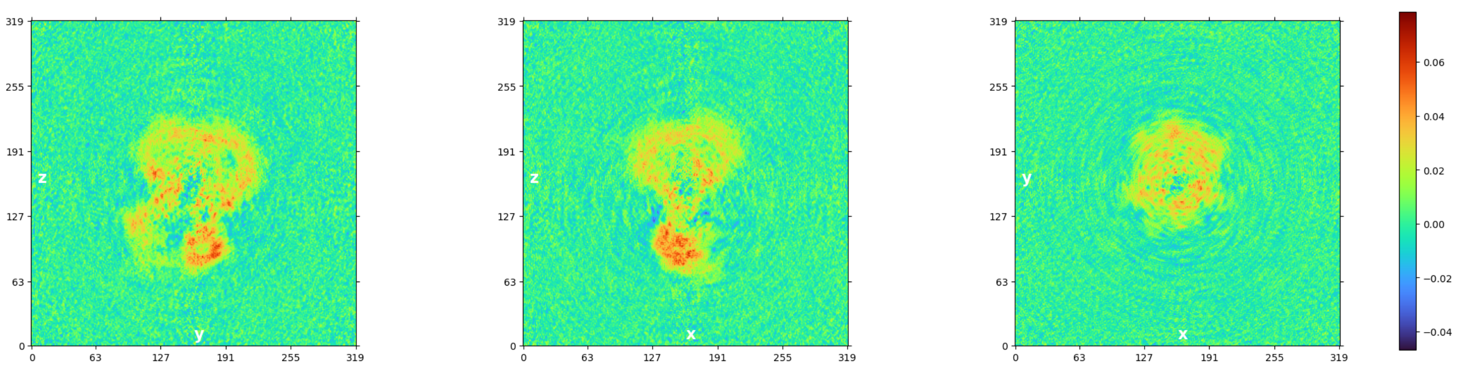


> [2026-06-09 15:36:06.35 GMT-4] Posterior Precision Directional Distribution Iteration 002

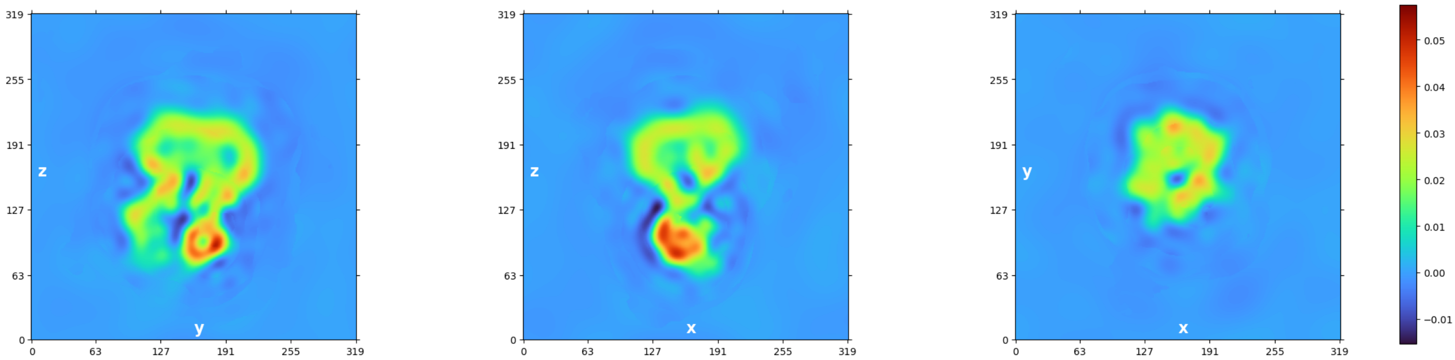



```
> [2026-06-09 15:36:06.35 GMT-4] [CPU: 10.72 GB]   [Avail: 295.60 GB]     Done in 18.823s.
> [2026-06-09 15:36:06.36 GMT-4] [CPU: 10.72 GB]   [Avail: 295.59 GB]   Outputting files..
> [2026-06-09 15:36:22.55 GMT-4] [CPU: 10.63 GB]   [Avail: 295.53 GB]     Done in 16.191s.
> [2026-06-09 15:36:22.56 GMT-4] [CPU: 10.63 GB]   [Avail: 295.53 GB] Done iteration 2 in 150.545s. Total time so far 433.561s
> [2026-06-09 15:36:23.90 GMT-4] [CPU: 10.63 GB]   [Avail: 295.92 GB] ---------------------------- Start Iteration 3
> [2026-06-09 15:36:23.90 GMT-4] [CPU: 10.63 GB]   [Avail: 295.92 GB]   Using Max Alignment Radius 29.506 (9.002A)
> [2026-06-09 15:36:23.90 GMT-4] [CPU: 10.63 GB]   [Avail: 295.92 GB]  Auto batchsize: 1515 in each split
> [2026-06-09 15:36:23.91 GMT-4] [CPU: 10.63 GB]   [Avail: 295.91 GB]   Using dynamic mask.
> [2026-06-09 15:36:54.91 GMT-4] [CPU: 10.63 GB]   [Avail: 289.73 GB] -- THR 1 BATCH 500 NUM 500 TOTAL 1.2861332 ELAPSED 12.658780
                                                                     --
> [2026-06-09 15:37:09.85 GMT-4] [CPU: 11.39 GB]   [Avail: 291.30 GB]   Processed 3023.000 images in 14.943s.
> [2026-06-09 15:37:13.05 GMT-4] [CPU: 11.39 GB]   [Avail: 290.95 GB]   Computing FSCs...
> [2026-06-09 15:37:13.06 GMT-4] [CPU: 11.39 GB]   [Avail: 290.93 GB]   Using full box size 320, downsampled box size 160, with low
                                                                     memory mode disabled.
> [2026-06-09 15:37:13.06 GMT-4] [CPU: 11.39 GB]   [Avail: 290.92 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:14.85 GMT-4] [CPU: 11.39 GB]   [Avail: 290.60 GB]     Done in 1.795s
> [2026-06-09 15:37:14.87 GMT-4] [CPU: 11.39 GB]   [Avail: 290.73 GB]   Optimizing FSC Mask...
> [2026-06-09 15:37:14.88 GMT-4] [CPU: 11.39 GB]   [Avail: 290.71 GB]   Using full box size 320, downsampled box size 160, with low
                                                                     memory mode disabled.
> [2026-06-09 15:37:14.89 GMT-4] [CPU: 11.39 GB]   [Avail: 290.71 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:17.79 GMT-4] [CPU: 11.64 GB]   [Avail: 291.27 GB]   Using full box size 320, downsampled box size 160, with low
                                                                     memory mode disabled.
> [2026-06-09 15:37:17.79 GMT-4] [CPU: 11.64 GB]   [Avail: 291.27 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:19.79 GMT-4] [CPU: 11.64 GB]   [Avail: 290.93 GB]   Using full box size 320, downsampled box size 160, with low
```

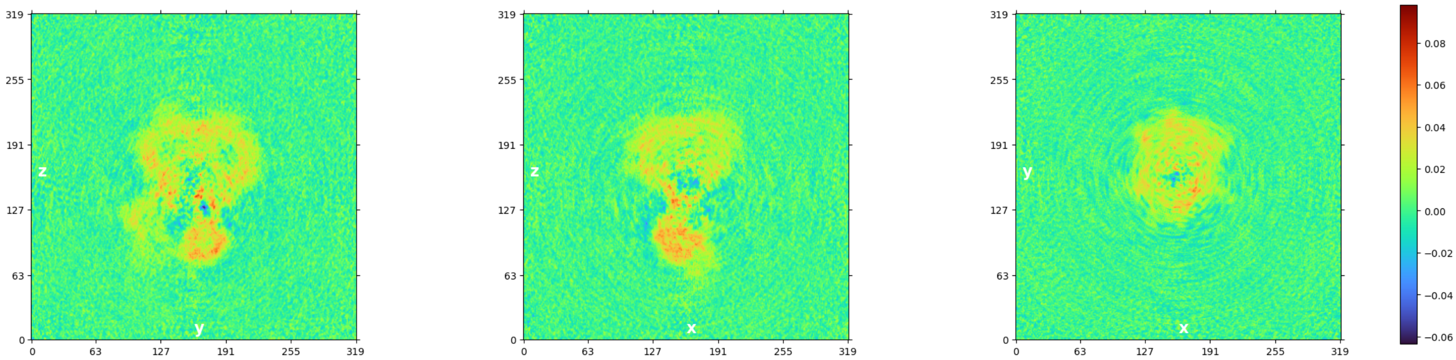
```
                                                                              memory mode disabled.
> [2026-06-09 15:37:19.79 GMT-4] [CPU: 11.64 GB]   [Avail: 290.93 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:21.87 GMT-4] [CPU: 11.64 GB]   [Avail: 291.54 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:21.87 GMT-4] [CPU: 11.64 GB]   [Avail: 291.53 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:23.93 GMT-4] [CPU: 11.64 GB]   [Avail: 291.34 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:23.94 GMT-4] [CPU: 11.64 GB]   [Avail: 291.34 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:25.89 GMT-4] [CPU: 11.64 GB]   [Avail: 290.65 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:25.90 GMT-4] [CPU: 11.64 GB]   [Avail: 290.64 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:27.92 GMT-4] [CPU: 11.64 GB]   [Avail: 288.70 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:27.93 GMT-4] [CPU: 11.64 GB]   [Avail: 288.70 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:30.00 GMT-4] [CPU: 11.64 GB]   [Avail: 287.51 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:30.00 GMT-4] [CPU: 11.64 GB]   [Avail: 287.51 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:32.18 GMT-4] [CPU: 11.64 GB]   [Avail: 288.44 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:32.19 GMT-4] [CPU: 11.64 GB]   [Avail: 288.44 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:34.19 GMT-4] [CPU: 11.64 GB]   [Avail: 288.36 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:34.20 GMT-4] [CPU: 11.64 GB]   [Avail: 288.36 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:36.30 GMT-4] [CPU: 11.64 GB]   [Avail: 288.28 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:36.30 GMT-4] [CPU: 11.64 GB]   [Avail: 288.28 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:38.25 GMT-4] [CPU: 11.64 GB]   [Avail: 287.57 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:38.26 GMT-4] [CPU: 11.64 GB]   [Avail: 287.57 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:40.22 GMT-4] [CPU: 11.64 GB]   [Avail: 287.45 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:40.23 GMT-4] [CPU: 11.64 GB]   [Avail: 287.45 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:42.26 GMT-4] [CPU: 11.64 GB]   [Avail: 288.50 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:42.27 GMT-4] [CPU: 11.64 GB]   [Avail: 288.50 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:44.18 GMT-4] [CPU: 11.64 GB]   [Avail: 289.35 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:44.18 GMT-4] [CPU: 11.64 GB]   [Avail: 289.35 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:46.05 GMT-4] [CPU: 11.64 GB]   [Avail: 289.31 GB]   Using full box size 320, downsampled box size 160, with low
                                                                              memory mode disabled.
> [2026-06-09 15:37:46.05 GMT-4] [CPU: 11.64 GB]   [Avail: 289.31 GB]   Computing FFTs on GPU.
> [2026-06-09 15:37:47.93 GMT-4] [CPU: 11.64 GB]   [Avail: 288.36 GB]     Done in 33.054s
> [2026-06-09 15:37:47.93 GMT-4] [CPU: 11.64 GB]   [Avail: 288.39 GB]   Computing cFSCs...
> [2026-06-09 15:37:49.96 GMT-4] [CPU: 11.64 GB]   [Avail: 287.40 GB]     Done in 2.022s
> [2026-06-09 15:37:49.96 GMT-4] [CPU: 11.64 GB]   [Avail: 287.40 GB]   Using Filter Radius 23.859 (11.132A) | Previous: 29.506
                                                                              (9.002A)
> [2026-06-09 15:37:56.51 GMT-4] [CPU: 10.88 GB]   [Avail: 287.59 GB]   Non-uniform regularization with compute option: GPU
> [2026-06-09 15:37:56.51 GMT-4] [CPU: 10.88 GB]   [Avail: 287.59 GB]   Running local cross validation for A ...
> [2026-06-09 15:38:20.64 GMT-4] [CPU: 11.13 GB]   [Avail: 290.14 GB]   Local cross validation A done in 24.128s
> [2026-06-09 15:38:21.97 GMT-4] FSC Filtered Side A
```



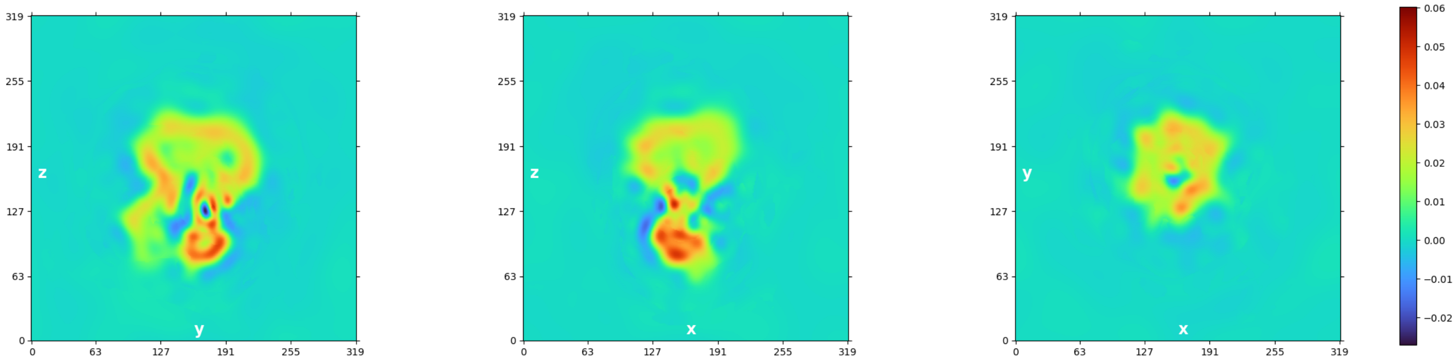
> [2026-06-09 15:38:22.92 GMT-4] CV Filtered Side A



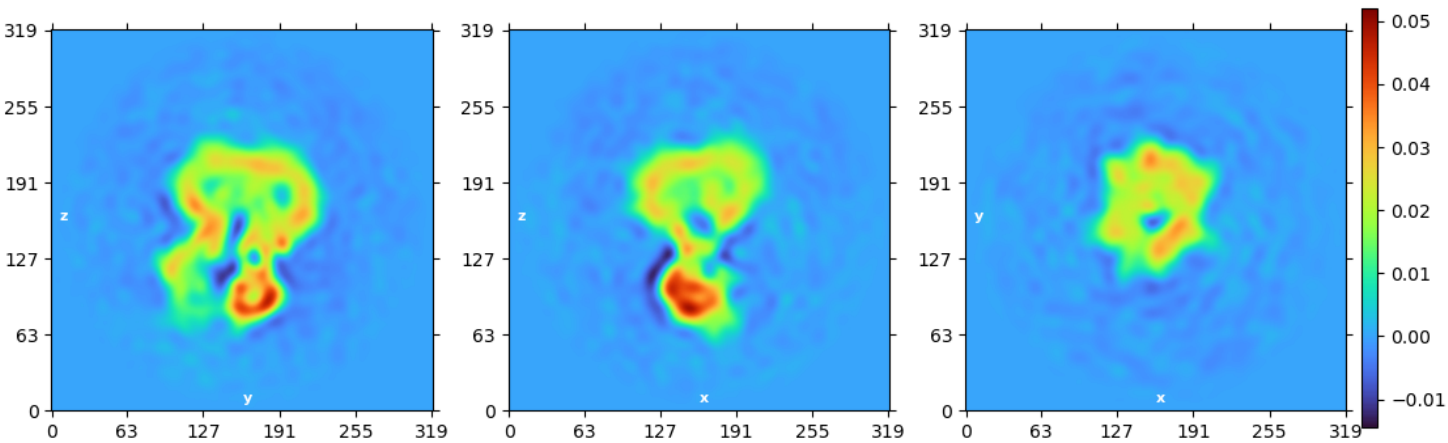
> [2026-06-09 15:38:22.92 GMT-4] [CPU: 11.13 GB]   [Avail: 289.59 GB]   Running local cross validation for B ...
> [2026-06-09 15:38:44.63 GMT-4] [CPU: 11.38 GB]   [Avail: 287.64 GB]   Local cross validation B done in 21.707s
> [2026-06-09 15:38:45.86 GMT-4] FSC Filtered Side B



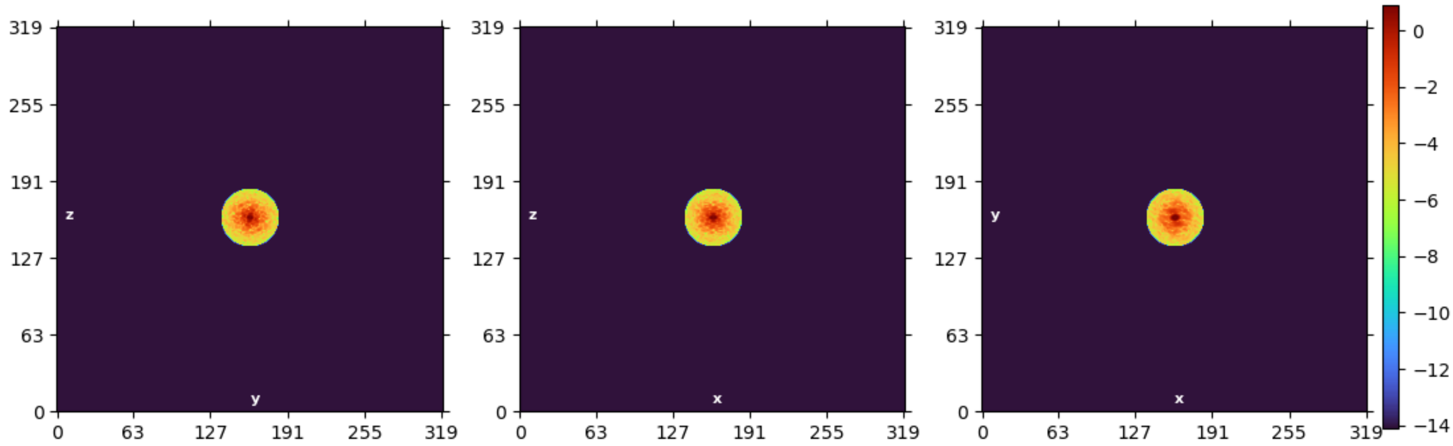
> [2026-06-09 15:38:47.00 GMT-4] CV Filtered Side B



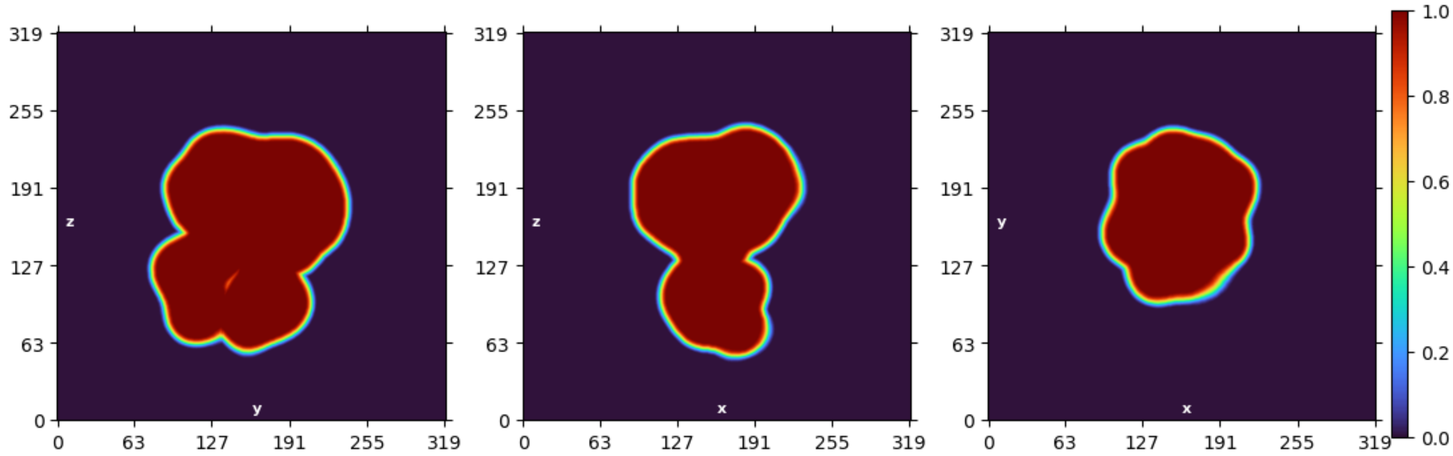
> [2026-06-09 15:38:54.53 GMT-4] [CPU: 11.26 GB]   [Avail: 288.76 GB]   Plotting..
> [2026-06-09 15:39:00.88 GMT-4] Real Space Slices Iteration 003



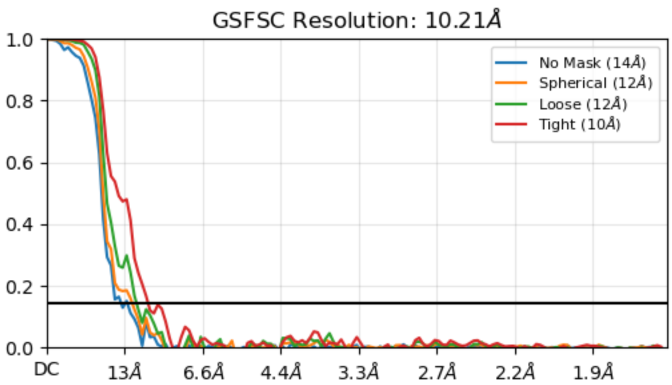
> [2026-06-09 15:39:02.28 GMT-4] Fourier Space Slices Iteration 003



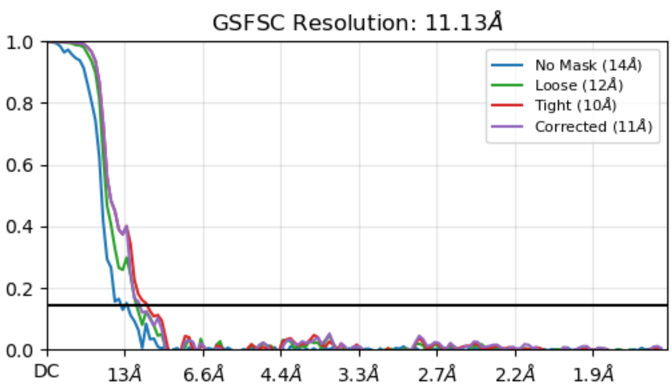
> [2026-06-09 15:39:04.26 GMT-4] Real Space Mask Slices Iteration 003



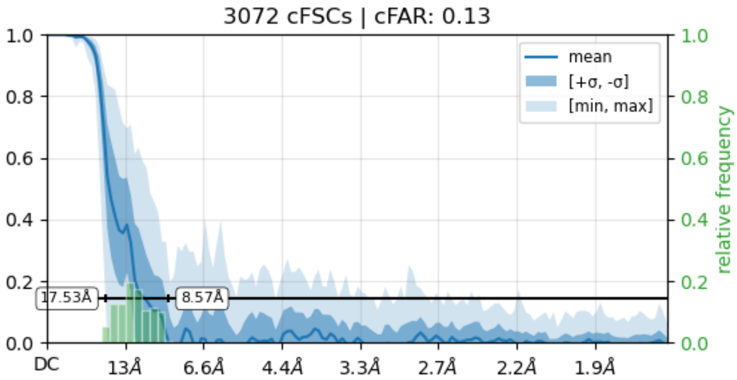
> [2026-06-09 15:39:04.74 GMT-4] FSC Iteration 003



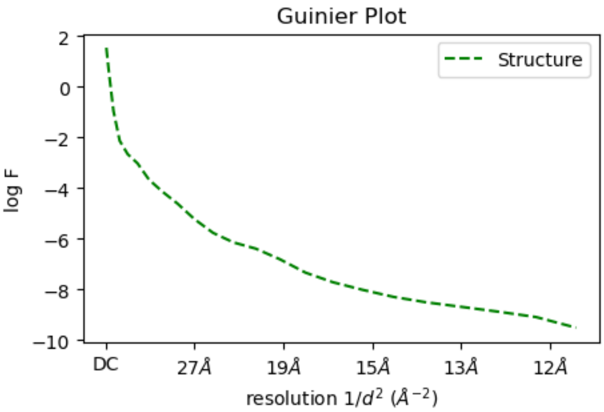
> [2026-06-09 15:39:05.22 GMT-4] FSC Iteration 003, after FSC-mask auto-tightening



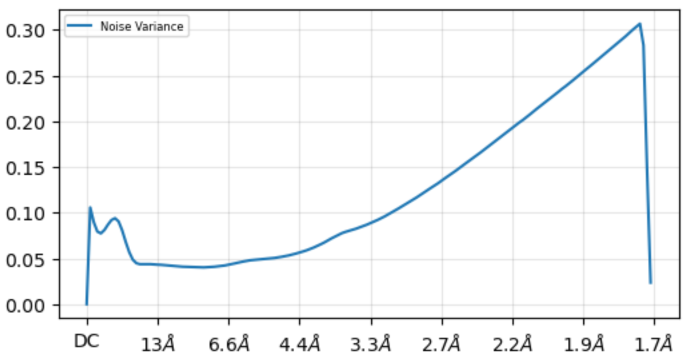
> [2026-06-09 15:39:05.93 GMT-4] cFSCs (Half-angle: 20°) Iteration 003, with autotight mask



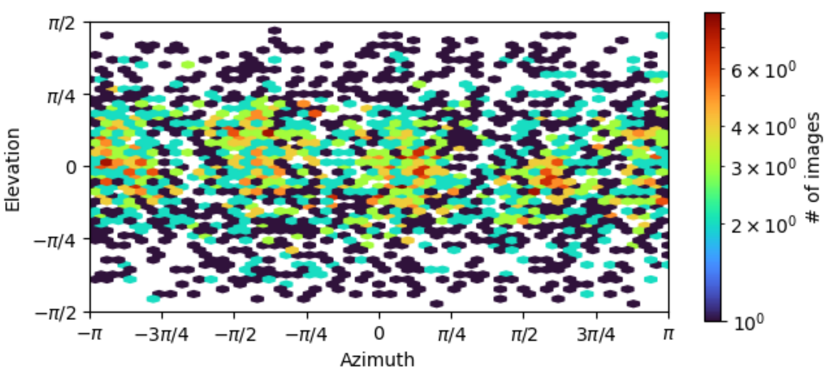
> [2026-06-09 15:39:11.64 GMT-4] Guinier Plot Iteration 003



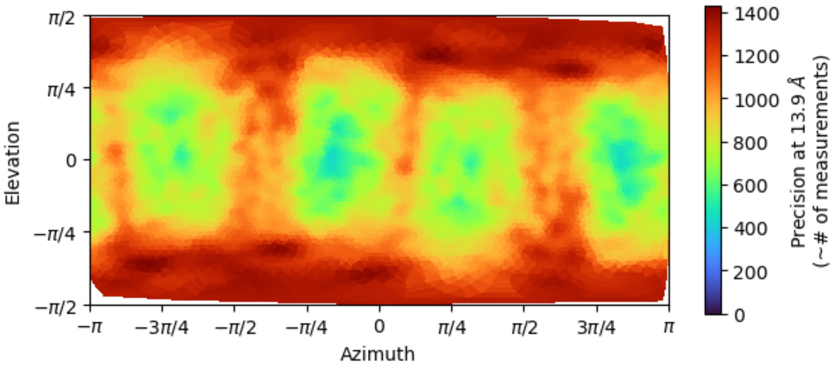
> [2026-06-09 15:39:12.11 GMT-4] Noise Model Iteration 003



> [2026-06-09 15:39:12.96 GMT-4] Viewing Direction Distribution Iteration 003



> [2026-06-09 15:39:14.80 GMT-4] Posterior Precision Directional Distribution Iteration 003



> [2026-06-09 15:39:14.80 GMT-4] [CPU: 11.32 GB] [Avail: 294.55 GB] Done in 20.268s.
> [2026-06-09 15:39:14.80 GMT-4] [CPU: 11.32 GB] [Avail: 294.55 GB] Outputting files..
> [2026-06-09 15:39:32.83 GMT-4] [CPU: 11.23 GB] [Avail: 301.78 GB] Done in 18.026s.
> [2026-06-09 15:39:32.84 GMT-4] [CPU: 11.23 GB] [Avail: 301.78 GB] Done iteration 3 in 188.935s. Total time so far 623.842s
> [2026-06-09 15:39:33.55 GMT-4] [CPU: 11.23 GB] [Avail: 301.75 GB] ====== Done Refinement ======
> [2026-06-09 15:39:33.56 GMT-4] [CPU: 11.23 GB] [Avail: 301.75 GB] Note that the output structure from refinement has been lowpass filtered to the gold-standard FSC resolution estimated
above. However, the structure probably needs sharpening in order to best visualize high resolution features. This can be

done with the Sharpening task (as a new experiment).
> [2026-06-09 15:39:33.56 GMT-4] [CPU: 11.23 GB] [Avail: 301.75 GB] Full run took 624.569s
> [2026-06-09 15:39:34.25 GMT-4] [CPU: 5.37 GB] [Avail: 307.45 GB] ---
> [2026-06-09 15:39:34.25 GMT-4] [CPU: 5.37 GB] [Avail: 307.44 GB] Compiling job outputs...
> [2026-06-09 15:39:34.26 GMT-4] [CPU: 5.37 GB] [Avail: 307.44 GB] Passing through outputs for output group particles from input group particles
> [2026-06-09 15:39:34.28 GMT-4] [CPU: 5.37 GB] [Avail: 307.41 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-09 15:39:34.29 GMT-4] [CPU: 5.37 GB] [Avail: 307.40 GB] Loaded output dset with 3023 items
> [2026-06-09 15:39:34.29 GMT-4] [CPU: 5.37 GB] [Avail: 307.40 GB] Passthrough results ['blob', 'alignments3D_multi']
> [2026-06-09 15:39:34.37 GMT-4] [CPU: 5.37 GB] [Avail: 307.33 GB] Loaded passthrough dset with 3023 items
> [2026-06-09 15:39:34.38 GMT-4] [CPU: 5.37 GB] [Avail: 307.33 GB] Intersection of output and passthrough has 3023 items
> [2026-06-09 15:39:34.40 GMT-4] [CPU: 5.37 GB] [Avail: 307.33 GB] Output dataset contains: ['blob', 'alignments3D_multi']
> [2026-06-09 15:39:34.41 GMT-4] [CPU: 5.37 GB] [Avail: 307.33 GB] Outputting passthrough result blob
> [2026-06-09 15:39:34.41 GMT-4] [CPU: 5.37 GB] [Avail: 307.33 GB] Outputting passthrough result alignments3D_multi
> [2026-06-09 15:39:34.42 GMT-4] [CPU: 5.37 GB] [Avail: 307.33 GB] Checking outputs for output group particles
> [2026-06-09 15:39:34.52 GMT-4] [CPU: 5.37 GB] [Avail: 307.24 GB] Updating job size...
> [2026-06-09 15:39:34.70 GMT-4] [CPU: 5.37 GB] [Avail: 307.06 GB] Exporting job and creating csg files...
> [2026-06-09 15:39:35.24 GMT-4] [CPU: 5.37 GB] [Avail: 305.68 GB] ***
> [2026-06-09 15:39:35.25 GMT-4] [CPU: 5.37 GB] [Avail: 305.68 GB] Job complete. Total time 626.29s